
Provably Bounded Autonomous Driving

A Collection of Ten Foundational Articles

Consolidated Project Corpus

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Constraint Propagation from Epistemology to Actuation: A Synthesis Toward Provably Bounded Autonomous Driving

Abstract

This synthesis advances a single thesis: that safety in autonomous driving should be engineered as an **unbroken chain of constraint propagation**, in which every degree of freedom available to a fallible perception-and-planning stack is narrowed, stage by stage, until unsafe actuator commands are not merely discouraged by policy but rendered physically unavailable at the actuator interface. The argument draws on four bodies of work supplied as primary material: a co-design safety stack, an emerging theory of neural-network learning dynamics, a survey of automotive sensing/numerics/functional-safety engineering, and a meta-level inventory of guarantee forms over a perception loop. It treats the mature, independently verifiable engineering and theory in that corpus as foundation, and it reframes the corpus’s more speculative architectural scaffolding as the candidate’s *proposed* contributions rather than as settled prior art. The result is a defensible spine for a doctoral program of work — together with an explicit account of which claims are established, which are proposed, and which could not be verified at all.

Note on sources and evidentiary standing

A dissertation lives or dies on the integrity of its citations, so this synthesis begins by being candid about the provided corpus. The four documents are uneven in their evidentiary footing, and an honest synthesis must say so before building on them.

What is well grounded. Document 3 (automotive computing and sensing) is largely a faithful survey of real engineering: MEMS inertial sensing and its vibration-rectification failure modes, event-driven dynamic-vision sensors, NV-center magnetometry and single-photon LiDAR detection, FMCW-radar FFT accelerator trade-offs, neuromorphic execution of spiking networks, RISC-V dual-core-lockstep paths to ASIL-D, Posit versus IEEE-754 stability in Kalman filtering, symplectic (Verlet) integration, AEC-Q100/ISO 16750-3 qualification, and V2X cooperative-perception attacks with trust-scoring defenses. Document 2’s underlying components are also real and well attested: the effective-field-theory program of Roberts, Yaida, and Hanin; Watanabe’s singular learning theory and the Local Learning Coefficient; the implicit-bias results associated with Srebro and collaborators; group-equivariant architectures; and sparse-autoencoder mechanistic interpretability. The control-theoretic core of Document 1 is likewise genuine: control-barrier functions, layered reduced-order/full-order control, recurrent tracking functions, Poisson safety functions, robust and neural CBFs, gameplay/adversarial safety filters, and the international standards landscape (ISO 26262, ISO 21448/SOTIF, UL 4600, RSS, SFF, rulebooks, IEEE 2846). The “Guarded Swarms” analog-veto principle — independent hardwired

guard channels that can unilaterally clamp an actuator command within microseconds, beneath any software fault — is a real and load-bearing idea drawn from a genuine publication.

What is proposed, not established. Several named frameworks that the corpus presents as if they were citable prior art do not appear in the literature and should be read as *constructed organizing devices*: “Constitutional Truth Governance,” the “Universal Filtration Primitive,” the “Trusted Scalar calculus,” the “Risk-Monotone Action Algebra,” and the “PICAPD” instruction-set architecture (with its specific 24-core die, 512-bit event-register file, and privileged `msafety` CSR). The umbrella term “Learning Mechanics” as a unified coalition discipline, and at least one citation it rests on, could not be verified. This synthesis does not discard these constructs — several are genuinely useful — but it relocates them. They are treated as **the dissertation’s own proposed contributions**: novel syntheses of real prior art, advanced as hypotheses to be defended, not as established results to be cited. Where a specific hardware artifact (the PICAPD ISA) is invoked, it is treated as a *notional reference architecture* — a design conjecture illustrating requirements, not a fabricated citation.

This relocation is not a weakness of the program; it is the program. A doctoral contribution is precisely a defensible new synthesis of established components. The discipline required is only that the seam between *what is known* and *what is claimed* never be blurred.

1.1 Problem statement and central thesis

Autonomous vehicles fail in a characteristic way. The hardware functions perfectly; the algorithm misreads a novel scene, mis-fuses a corrupted sensor stream, or — increasingly — hallucinates under a learned policy whose behavior on the present input was never directly tested. Classical functional safety (ISO 26262) was built for the opposite failure: it assumes the *hardware* breaks and the specification is correct. The gap between these two pictures is the territory that ISO 21448 (SOTIF) names “the unknown unsafe,” and it is the territory in which autonomous driving actually operates.

The conventional response is to make the perception-and-planning stack more accurate. The thesis of this work is that accuracy is the wrong target. **A stack can be made arbitrarily accurate on average and still emit a catastrophic command on the one input that matters.** The alternative is to make the *consequences* of error bounded by construction: to accept that the high-level stack is fallible, and to interpose a chain of constraints — epistemic, numerical, control-theoretic, and finally electrical — such that the worst an error can do shrinks monotonically as it travels toward the wheels.

Stated formally, the thesis is a claim about monotonicity. Define, at each stage of the loop, the set of actuator commands the stage is permitted to authorize. The central claim is:

As one moves from perception toward actuation, and as estimated risk rises, the admissible-command set may only contract, never expand — and at the final, pre-actuation stage this contraction is enforced by physics rather than software.

Everything that follows is an attempt to make that claim precise, to ground each link in established theory, and to identify where the link is currently a proposal rather than a proof.

1.2 Foundations accepted from the literature

This chapter records the established results the dissertation builds on. None of it is original; all of it is load-bearing.

1.2.1 Layered control and the reduced-order/full-order gap

A high-dimensional vehicle (a legged platform’s 36-dimensional state, or a tracked vehicle’s relative-degree pathologies) cannot have control-barrier functions synthesized on its full dynam-

ics under microsecond deadlines. The accepted resolution is *layered control*: synthesize the CBF on a low-dimensional reduced-order model (RoM) capturing the safety-critical kinematics, and task a low-level controller with forcing the full-order model (FoM) to track the RoM trajectory. The safety guarantee is only as good as the tracking bound. The relevant advance is the **recurrent tracking function (RTF)**, which relaxes strict monotonic Lyapunov decay — available systematically only for fully actuated systems — to a weaker finite-time recurrence condition, permitting transient tracking error provided the trajectory returns to a recurrent zero-superlevel set before leaving the safe region. RTFs are what let a guarantee proven cheaply on the RoM project onto the physical FoM.

1.2.2 Robustness, adaptation, and the cost of conservatism

Real state estimation drifts; visual-inertial odometry accumulates error between global fixes. Constant worst-case robustness margins (standard robust CBFs) over-tighten until the optimization is infeasible. The accepted refinement is **optimization-based online parameter adaptation** of the robustness margin, combined with **Poisson safety functions** — solving Poisson’s equation with Dirichlet boundary conditions on the safe/unsafe sets to merge many constraints into a single smooth scalar field that resolves quickly. **Neural CBFs and game-play filters** (adversarial self-play between a safety-seeking policy and a worst-case-uncertainty adversary) internalize these constraints offline for zero-shot robustness. These are the credible state of the art in tractable safety filtering.

1.2.3 Sensing and numerics as the substrate of trust

A safety margin is only as honest as the numbers feeding it. The corpus supplies the relevant facts: MEMS inertial noise floors and vibration-rectification offsets set a hard lower bound on how small a real “awakening” threshold can be; event-driven vision and single-photon LiDAR change the latency/dynamic-range envelope; and — most consequentially for any safety scalar — **the numerical format is a safety variable**. Standard IEEE-754 single precision can drive an extended Kalman filter to diverge after several hundred seconds as the covariance loses positive-definiteness; Posit arithmetic with quire accumulation can track a double-precision reference across the full horizon. The implication the dissertation takes forward: any scalar that gates actuation must be computed in a format whose rounding error is bounded on the *safe* side.

1.2.4 Symplectic prediction

Trajectory prediction over a horizon must not inject energy. Symplectic integrators (Störmer/Velocity-Verlet) bound the simulated energy to oscillate about its true value, with global error of order $O(\Delta t^2)$, avoiding the unphysical “teleportation” of non-symplectic schemes. This is the accepted basis for the predictive step of the loop.

1.2.5 The learning theory that makes the policy legible

Document 2’s genuine content is that deep networks need no longer be treated as black boxes. Two results matter here. First, **singular learning theory** replaces parameter count with the **Local Learning Coefficient (LLC)** as the true measure of effective complexity, estimable at scale via preconditioned stochastic-gradient Langevin dynamics. Second, training is **stagewise**: capabilities appear in discrete developmental phase transitions, visible as a sharp regret drop paired with an LLC jump. The dissertation uses these not as decoration but as instrumentation — a way to *detect*, online, that a learning policy has changed what it knows.

1.2.6 The standards landscape and the analog-veto principle

ISO 26262 governs electrical/electronic faults via the ASIL framework; SOTIF (ISO 21448) and UL 4600 address performance-limitation hazards with no hardware fault; RSS, SFF, rulebooks, and IEEE 2846 operationalize the behavioral “duty of care.” And from the Guarded Swarms work comes the principle that ties the stack to ground: digital coordination (there, Petri-net-based) negotiates *what to do*, but **independent analog guard channels at the actuator interface can veto an unsafe command within microseconds, without programmable logic in the guard path**. Intelligence proposes; hardware disposes.

1.3 Proposed contributions (the original synthesis)

This chapter advances the dissertation’s own claims. Each is a synthesis of the Chapter 2 foundations; none should be read as pre-existing literature. They are stated as proposals to be defended, with their anti-patterns and their verification obligations made explicit.

1.3.1 Contribution I — The fidelity gate as a *trusted scalar*

Proposal. Reconstruction fidelity in a perception loop should not be measured as reconstruction error (an L2 distance between input and rebuild). It should be a single qualified scalar φ that answers a different question: *can the discarded detail only ever shrink, never enlarge, the admissible action set?* Concretely, φ is required to satisfy a five-part contract — it must denote a genuine safety-relevant quantity (a barrier-margin or causal-sufficiency witness, not a PSNR proxy); remain bounded and saturating; **overestimate** risk whenever it errs; feed only constraint-*tightening* maps; and be re-qualified continuously, transitioning from trusted to suspect and escalating monotonically toward maximum risk on any conformance failure.

The inversion that makes it original. Classical estimation minimizes average two-sided error. This proposal deliberately bounds *worst-case optimistic* error and compiles that bound into an additive conservative bias, so the reported margin globally lower-bounds the true margin. A fast, fallible estimator becomes trustworthy not by being right but by being *wrong only in the safe direction*.

Anti-patterns it forbids. Treating L2 accuracy as the safety criterion; letting a “high-fidelity” reading authorize a more aggressive action; and — critically — **averaging**. Combining a trusted scalar with an untrusted one, or taking the less-conservative of two, destroys the worst-case bound. The admissible algebra is restricted to the conservative join and safe-monotone composition.

Verification obligation. This is the contribution most in need of grounding. It requires (a) a proof that the chosen φ for a given safe set is in fact a valid one-sided bound under stated assumptions, and (b) a numerical-format argument (Chapter 2.3) that the computation of φ cannot round in the optimistic direction. Without both, the contract is aspirational.

1.3.2 Contribution II — A risk-monotone actuation algebra

Proposal. At the boundary immediately before actuation, define a family of admissible-command sets indexed by a normalized risk band, with the **antitone** property that rising risk only shrinks the set. Require every admissible set to be a closed, convex, axis-aligned box, so that projecting any proposed command onto it is a unique, **firmly non-expansive (1-Lipschitz)** operation — guaranteeing the enforcer can never amplify an oscillation injected by corrupted software. Implement the projection as a two-stage clamp (magnitude, then rate) with the **rate clamp applied last**, so that a sudden risk spike produces bounded jerk (a graceful “distance-decay” toward the new envelope) rather than an actuator step-lurch.

Why the ordering is the contribution. The non-obvious, defensible claim is that *clamp ordering is a safety property*. Magnitude-last ordering forces an unbounded step when a risk

spike shrinks the box below the current state; rate-last ordering ratchets the command into the tighter envelope at maximal bounded jerk. The corpus’s worked example — a Band-4 transition bleeding steering and acceleration down over $\sim 1.25\text{--}1.47$ s rather than snapping — is the intuition pump; the dissertation owes the **exactness theorem** that, for axis-aligned constraints, sequential clamping equals the true Euclidean projection.

Composition and hysteresis. When multiple envelopes apply, the effective envelope is their **intersection** (meet) — never union or average. Band switching uses **asymmetric hysteresis**: tighten instantly on crossing an up-threshold, loosen only below a strictly lower down-threshold after a dwell time. Symmetric hysteresis is rejected because it delays tightening, which is the one direction that must never be delayed.

1.3.3 Contribution III — A timing-contracted analog veto

Proposal (grounded in real prior art). The actuation algebra of 3.2 is enforced by an independent analog layer under an explicit timing contract: intervention latency $\ll$ sensor-consistency window $\ll$ digital decision window. This is the dissertation’s adaptation of the Guarded Swarms analog-veto principle to the on-road single-vehicle case. Because the guard path is hardwired comparators and latches with no programmable logic, a compromised or hallucinating software stack cannot reach the actuator before the guard acts. On corrupted data, a missed deadline, or a tampered signal path, the guard forces the minimal-risk envelope (maximal safe braking, zero positive acceleration) — a **fail-closed** default rather than last-good-command coasting.

The notional hardware. The corpus’s “PICAPD ISA” is best read here as a *requirements sketch* for such a layer — privileged, non-user-writable safety registers; deterministic bounded latency; architectural rollback to a known-safe configuration on a violated safety bit. The dissertation should not cite it as an existing chip; it should state these as the properties any conforming enforcer must exhibit and note that FPGA-accelerated kernels are the credible near-term substrate for the jitter-bounded reflexive layer.

1.3.4 Contribution IV — A conformance checklist as the integrating artifact

Proposal. The contributions above compose into a single fail-closed release gate: a reconstruction may drive actuation only if, simultaneously, (1) φ is a qualified trusted scalar; (2) every fused input passed a *pre-fusion* admissibility test and carries provenance; (3) the RoM→FoM tracking error is within its recurrent bound; (4) the command lies in (or is non-expansively projected into) the intersection of all active envelopes, rate-clamp last; (5) no averaging, union, or less-conservative-meet appears anywhere on a safety-bearing path; (6) the retrieval that produced the reconstruction respected its compute/latency budget; (7) no online policy update silently revised ground truth; and (8) any failed check has already placed the system in the minimal-risk envelope. The items are conjunctive: the *absence* of a discharged guarantee is itself a fault.

This checklist is the dissertation’s claim to systematicity — it is what turns a collection of mechanisms into an argument that “the unknown unsafe is structurally bounded rather than empirically sampled.”

1.4 The learning-theory bridge: governing an updating policy

A vehicle whose policy updates online raises a governance problem the actuation layer cannot solve: the policy may *quietly* acquire a new behavior. Here the dissertation joins the two halves of the corpus.

Detector (established). The LLC and stagewise-transition results give a real, online-estimable signal: a developmental phase transition is observable as a coupled regret-drop /

LLC-jump. The singular-MDL result further bounds how far decomposition can compress an input before it destroys causally necessary structure — a principled floor on lossy perception.

Governance (proposed). The dissertation’s overlay is an **anti-silent-drift** rule: any update that alters the policy’s effective truth judgments must be recorded as a formal, challengeable event rather than absorbed into the canon. The detector fires; the governance logs and, where warranted, quarantines. This is where the corpus’s “constitutional” language is useful as a *design metaphor* — separation between a component that *proposes* a reconstruction and an independent one that can *refuse to canonize* it, with ambiguity preserved when evidence is insufficient. It should be presented as an architectural pattern (proposer/verifier/adjudicator separation), not as an established governance theory.

A genuine open risk: implicit bias as a double-edged sword. The corpus’s most useful caution from learning theory is that the optimizer’s implicit geometric bias — the very thing that makes overparameterized networks generalize — can *degrade* robustness when it is misaligned with the adversarial threat model’s geometry (the “price of implicit bias”). For a safety-critical policy this means: a network that generalizes beautifully on clean data may be systematically fragile in exactly the direction an attacker or a rare scene exploits. The dissertation should treat alignment between optimizer bias and the safety threat geometry as a first-class design constraint, not an afterthought.

1.5 Standards crosswalk and the certification argument

The payoff of the constraint chain is a certification posture that empirical road-miles alone cannot reach.

Table 1.1: Standards crosswalk

Standard	What it demands	Where the synthesis answers it
ISO 26262 (ASIL)	Mitigate E/E faults; freedom from interference	Hardware veto, deterministic bounded latency, lockstep substrate, numerical resilience
ISO 21448 / SOTIF; UL 4600	Bound the “unknown unsafe” with no hardware fault present	Trusted-scalar fidelity gate + actuation algebra make unsafe commands physically unrepresentable
IEEE 2846 / RSS / SFF	Minimum reasonable assumptions; behavior across ODD transitions	Rate-clamp-last bounded-jerk entry into a stricter RSS envelope; asymmetric hysteresis
AEC-Q100 / ISO 16750-3	Thermomechanical and vibration survivability	Vibration/noise floor sets the real lower bound on any safety threshold; format/precision choices

The decisive claim. The argument the chain licenses is narrow but strong: *even under severe algorithmic hallucination in perception or planning, the actuation-boundary forms force the vehicle into a provable Minimum Risk Condition*. This converts SOTIF compliance from a statistical claim about how rarely the stack errs into a structural claim about what an error can do — and it does so without violent braking at ODD transitions, because the same rate-clamp-last discipline that prevents lurching is what carries the vehicle smoothly into the more conservative RSS parameters that fog or rain demand.

1.6 Critical appraisal, limitations, and what must be built

A synthesis is only honest if it states where it could fail.

- **The exactness theorem is owed, not proven.** Contribution II’s claim that sequential magnitude-then-rate clamping equals Euclidean projection holds for axis-aligned convex boxes. Real actuator constraints (coupled steering/acceleration limits, non-box friction circles) are not axis-aligned. The dissertation must either prove the theorem extends to the relevant non-box geometry or restrict its claims to the regime where the box approximation is itself conservative.
- **The trusted scalar can be vacuously safe.** A scalar that always reports maximum risk satisfies every clause of the contract and parks the vehicle forever. The hard part — entirely unaddressed by the contract as stated — is *liveness*: proving the gate is not so conservative that it forbids ordinary driving. The genuine research problem is the trade-off frontier between the worst-case bound and feasibility, exactly the tension the robust-CBF adaptation literature already grapples with. This is the dissertation’s central empirical question, and it is open.
- **Detection latency versus actuation latency.** The learning-theory detector (LLC estimation via SGLD) operates on training-time or slow-loop timescales; the actuation veto operates in microseconds. The governance overlay therefore cannot prevent a hallucinated command in real time — it can only quarantine a *capability* between drives. The honest claim is that the analog veto is the real-time guarantee and the learning governance is a slower assurance layer; conflating their timescales would overstate the result.
- **The notional hardware must not masquerade as built.** The “PICAPD” architecture and its register-budget arithmetic are useful as a requirements illustration. Presenting specific cycle counts, opcodes, or register files as measured properties of an existing device would be a citation failure. The defensible version states the *required* properties (privilege isolation, bounded rollback latency, jitter-bounded execution) and points to FPGA/lockstep-RISC-V as the realizable substrate.
- **Unverified citations must be removed or downgraded.** At least one source the corpus leans on (the “Learning Mechanics” coalition framing, and an associated preprint identifier) could not be confirmed. Before submission, every citation on a load-bearing path must be independently verified; unverifiable ones should be cut or explicitly flagged as the candidate’s own conjecture.

1.7 Conclusion

The contribution of this synthesis is not a new optimizer, sensor, or controller. It is an *architecture of refusal*: a disciplined chain in which each stage can only narrow what the next is permitted to do, terminating in an analog layer that enforces the narrowing with electricity rather than code. The foundations of that chain — layered CBF/RTF control, robust and learned safety filters, the sensing and numerical substrate, the singular-learning instrumentation that makes a policy legible, and the analog-veto principle of Guarded Swarms — are real and cited as such. The integrating mechanisms — the trusted-scalar fidelity gate, the risk-monotone actuation algebra, the timing-contracted veto, and the conformance checklist that binds them — are advanced here as the candidate’s own proposals, with their open obligations (an exactness theorem beyond box constraints, a liveness/feasibility frontier, a real-time/quarantine timescale separation, and a notional-versus-built hardware boundary) stated plainly.

The wager of the program is that this is the right level of abstraction at which to make autonomous driving *certifiable* rather than merely *tested*: not by proving the stack never errs, but by proving that when it does, the error has nowhere unsafe to go.

Appendix — Provenance ledger

Table 1.2: Provenance ledger

Claim or construct	Standing	Treatment in this synthesis
CBF / RoM–FoM layered control / RTF / Poisson safety / neural & gameplay filters	Established	Accepted as foundation (Ch. 2.1–2.2)
MEMS, DVS, quantum sensing, FFT accelerators, neuromorphic, RISC-V ASIL-D, Posit, Verlet, AEC-Q100	Established	Accepted as substrate (Ch. 2.3–2.4)
Principles of Deep Learning Theory; SLT/LLC; implicit bias; equivariance; SAEs/mech-interp	Established	Accepted; LLC used as detector (Ch. 2.5, 4)
ISO 26262, SOTIF/21448, UL 4600, RSS, SFF, rulebooks, IEEE 2846	Established	Accepted; certification cross-walk (Ch. 5)
Guarded Swarms analog-veto principle (TB-CSPN + microsecond actuator vetoes)	Established (verified)	Accepted; basis of Contribution III
“Trusted Scalar” five-part contract	Not in literature	Repositioned as Contribution I (proposal)
“Risk-Monotone Action Algebra” / clamp-ordering	Not in literature	Repositioned as Contribution II; exactness theorem owed
“Constitutional Truth Governance” / “Universal Filtration Primitive”	Not in literature	Used as design metaphors only (Ch. 4)
“PICAPD” ISA (24-core die, 512 event regs, <code>msafety</code> CSR)	Could not be verified	Treated as notional requirements sketch
“Learning Mechanics” coalition umbrella + associated preprint ID	Could not be verified	Downgraded; underlying components cited individually

Invariants Before Identities: A Provenance-Disciplined Synthesis of the Elliptic-Integral Surrogate Framework

Abstract

This synthesis advances a single thesis: that the durable contribution of the elliptic-integral / AGM framework is **not** any single striking correspondence it surfaces — least of all the numerical proximity of its gain to cosmological-constant scales — but a *constructive, conservation-respecting surrogate* that emits two cheap, interpretable invariants (a continuous statistic ξ and a discrete index S) whose claims are admissible only at the level the data licenses. The argument draws on the project corpus supplied as primary material: the dissertation manuscript and its enhanced revision, a unified-framework consolidation paper, four “support” addenda (SR1–SR4) that re-state the method as calibrated hypotheses, and an exploratory biological-bridge analysis. It treats the mature numerics and the bin-level empirical findings as foundation, and it reframes the corpus’s more speculative scaffolding — the cosmological identity, the perfect-fluid convergence law, the cross-domain “robust bridge” to biology — as the candidate’s *proposed* contributions, each carrying an explicit verification obligation, rather than as settled results. The organizing principle is a monotonicity discipline borrowed from the project’s own stated ambition: **claim-strength may rise only as verification obligations are discharged, and may never run ahead of its evidence.** The “robust perfect” the program seeks is precisely the fixed point of that vetting process.

Note on sources and evidentiary standing

A dissertation lives or dies on the integrity of its claims, so this synthesis begins by being candid about the corpus. The provided documents are uneven in their evidentiary footing — and, importantly, *uneven with themselves across revisions*. An honest synthesis must say so before building on them.

What is well grounded. The numerical core is real and classical. The complete elliptic integral of the first kind, evaluated by the arithmetic-geometric mean, converges quadratically — roughly doubling correct digits per iteration, reaching machine precision in about five steps — and this is textbook numerics with a long heritage in celestial mechanics. The elliptic-filter realization (the `ellipap` route) returning a factored transfer function $H(s) = k \prod_i (s - z_i) / \prod_j (s - p_j)$ is standard control engineering. The *empirical* findings on the gasification dataset are also genuine and, in the later addenda, carefully scoped: mass closure is achieved on the tested cases without explicitly solving the governing PDEs; the dimensionless log-deviation $\xi = \ln(\Lambda_S/\Lambda_G)$ is monotone with binned product-gas yield; the Squarity index $S = |m - n| \in \{0, 1\}$ is monotone with — and “locks” at high — binned carbon conversion; and `sct` covaries with non-equilibrium

stiffness across cases. The canonical-correlation compaction of biomass properties into a small composite is a real statistical procedure. Most consequentially, the **bin-level validation discipline** stated in SR2–SR4 is itself a methodological strength: the corpus explicitly declines to claim individual-sample validity and reports group-level monotonicity with confidence summaries.

What is proposed, not established. Several constructs the corpus at times presents as discoveries are better read as *candidate contributions advanced as hypotheses*: the treatment of the gain k as a physical constitutive constant identifiable with the cosmological term Λ (the “ $\Lambda(x) = \Lambda_{Ge^{\xi(x)}}$ ” mapping); the perfect-fluid convergence law $\varepsilon \equiv \text{median}|\xi| + \alpha \text{Var}(\text{sct}) + \beta \mathbf{1}[S=1] \rightarrow 0$; the assertion that two correlates constitute “non-trivial validation”; and the domain-agnostic transfer of (ξ, S) to metabolic and signaling networks. The strongest form of each appears in the earlier register — **enhanced_thesis**’s claim that the cosmological alignment is “*not coincidental*,” and **invariant_bridge_analysis**’s verdict that “the Bridge is Robust” with 10^5 – $10^6 \times$ speedups — and the calibrated later register (SR2–SR4) already walks these back to “heuristic cue,” “operational normalization,” and “conjecture.” This synthesis completes that walk-back and makes it structural.

What could not be verified. The unmeasured speedup figures (computed against a hypothetical ODE baseline, not a benchmarked solver), the cross-scale claim that “the same objects ξ, S appear at every biological scale,” and the open-stance gestures toward quantum-information correspondence are not supported by anything in the corpus that an examiner could check. They are downgraded to motivation, not retained as results.

This relocation is not a weakness of the program; **it is the program**. The project’s own note states the ambition exactly — “we are not worried if our early *perfect* invariant bridge analysis could not withstand interrogation... we focus on the design of vetting procedures that’ll get us there.” The discipline required is only that the seam between *what is observed*, *what is proposed*, and *what is merely evocative* never be blurred.

2.1 Problem statement and central thesis

Surrogate models of complex physical systems fail in a characteristic way. The numerics are sound; the *interpretation* over-reaches — a fit that holds on a tested cohort is narrated as a law, a dimensional coincidence is narrated as a mechanism, a within-domain regularity is narrated as a universal bridge. High-fidelity CFD is faithful but unit- and fuel-specific; equilibrium black-box models are broad but blind to non-equilibrium structure; machine-learning surrogates are flexible but routinely violate the conservation laws they are meant to respect. The framework targets the genuine middle ground: a construction that tolerates feedstock heterogeneity, respects mass and energy closure by its transfer-function structure, surfaces interpretable internal invariants, and yields fast bin-level inference.

The thesis of this work is that the middle ground is won or lost not on accuracy but on *claim discipline*. **A surrogate can be empirically excellent on a cohort and still mislead catastrophically the moment its narration outruns its evidence** — when a per-domain normalization is sold as a physical identity, or a 200-case monotonic trend is sold as single-sample predictive power. The alternative is to make claim-strength *bounded by construction*: to accept that the surrogate is an approximation, and to interpose a ladder of verification obligations — numerical, statistical, physical, and cross-domain — such that the strength of any assertion can only rise as the corresponding obligation is discharged.

Stated formally, the thesis is a claim about monotonicity. Define, at each rung of the ladder, the set of assertions the evidence currently licenses. The central claim is:

As one moves from raw numerics toward physical and cross-domain interpretation, the set of *licensed* claims may expand only when a stated verification obligation is discharged — and the framework’s headline correspondences (cosmological, biological,

information-theoretic) sit at rungs whose obligations are, as yet, **undischarged**.

Everything that follows makes that ladder explicit: grounding each established rung, naming each proposed rung as a proposal, and stating for each the precise obligation that would license its promotion.

2.2 Foundations accepted from the corpus

This chapter records the established results the synthesis builds on. None is in dispute; all are load-bearing.

2.2.1 The AGM evaluation and its convergence

The arithmetic-geometric-mean evaluation of the complete elliptic integral of the first kind, with the standard triple $a_0 = 1$, $b_0 = \sqrt{1-m}$, $c_0 = \sqrt{m}$ and the recursions $a_i = \frac{1}{2}(a_{i-1} + b_{i-1})$, $b_i = \sqrt{a_{i-1}b_{i-1}}$, $c_i = \frac{1}{2}(a_{i-1} - b_{i-1})$, converges quadratically. This is the framework’s most secure asset: a numerically stable, well-characterized integrator with a deep heritage. The *constructive* output — explicit zeros, poles, and gain — is directly inspectable, which is what makes downstream interpretability possible at all.

2.2.2 The transfer-function realization and conservation by structure

Routing the integration through an elliptic low-pass filter prototype yields a proper rational $H(s) = k \prod_i (s - z_i) / \prod_j (s - p_j)$. The accepted, defensible benefit of this representation is *structural*: a proper, stable rational map encodes passivity, stability (poles in the open left half-plane), and causality — properties that a generic neural surrogate can and does violate (negative concentrations, mass created from nothing). The honest statement is that the **structure forbids a class of conservation violations by construction**; this is real and is the framework’s strongest claim to “physics-respecting” inference.

2.2.3 The two invariants, precisely defined

$\xi \equiv \ln(\Lambda_S/\Lambda_G)$, with $\Lambda_S \equiv k$ the per-sample gain and Λ_G the cohort geometric mean, is a dimensionless, signed, two-tailed log-deviation. $S \equiv |m - n| \in \{0, 1\}$ (“Squarity”) is the degree-balance of the realization. Both are well-defined functions of the $H(s)$ realization and require no external tuning. Their definitions are clean and their computation is cheap; this is not in question.

2.2.4 The bin-level empirical findings

On the 200+ gasification cases: mass closure is observed without explicit PDE solves; ξ is monotone with binned yield; S is monotone with (and locks at high) binned conversion; **sct** covaries with non-equilibrium stiffness; and a 72% correlation links input and output geometric ratios. These are the framework’s empirical anchors, and they are reported at the correct grain.

2.2.5 The validation discipline itself

The corpus’s most underrated established asset is its *epistemics*. SR2–SR4 state plainly that individual samples are not claimed validated; that inference is at the bin level; that S is a deterministic per-case class, not a dynamic “flip”; that the cosmological link is a heuristic cue, not an identity; and that elevation to individual-level prediction would require an order-of-magnitude more data under controlled, pre-registered perturbation. This calibrated stance is foundation, not decoration — it is the part of the corpus that already survives interrogation.

2.3 Proposed contributions (the original synthesis)

This chapter advances the framework’s own claims. Each is a synthesis of the Chapter 2 foundations; none should be read as established. Each is stated with the anti-pattern it must avoid and the verification obligation that would license its promotion.

2.3.1 Contribution I — The invariant pair as an orthogonal, falsifiable validation instrument

Proposal. (ξ, S) should be treated not as two more fitted features but as a *paired instrument*: a continuous lever (ξ , against yield) and an independent discrete lever (S , against conversion). The claim to originality is that aligning two *independent* internal invariants with two *independent* external observables is harder to fake than a single regression and therefore constitutes stronger evidence than an R^2 .

The inversion that makes it original. Classical surrogate validation maximizes average fit on one target. This proposal instead demands *simultaneous monotone alignment on two orthogonal axes from invariants the model did not see* — turning validation from a curve-fit into a structural prediction.

Anti-patterns it forbids. Reporting ξ and S as if they were independently established when they are both functions of the same $H(s)$; treating bin-level monotonicity as individual-sample predictive power; and choosing the S -proxy post hoc to fit the data.

Verification obligation. (a) A statistical demonstration that ξ and S are genuinely *orthogonal* (not two readouts of one latent), e.g. via partial-correlation or ablation showing each carries signal the other does not; (b) the SR3/SR4 ablation falsifier actually run — removing `sct`, randomizing the embedding, dropping CCA — and shown to degrade closure and the (ξ, S) ties; (c) pre-registration of the binning scheme to foreclose researcher degrees of freedom. Until (a)–(c) are discharged, the instrument is *promising*, not *proven*.

2.3.2 Contribution II — The constitutive normalization $\Lambda(x) = \Lambda_G e^{\xi(x)}$ as an operational scale-free field

Proposal. Within the tested domain, the gain may be normalized into a scale-free field $\Lambda(x) = \Lambda_G e^{\xi(x)}$ that behaves consistently with observed yield variation. This is the *defensible residue* of the cosmological excursion: an operational normalization, explicitly bounded to the domain.

Why the demotion is the contribution. The non-obvious, defensible move is to *refuse* the identity. The numerical proximity of k to Λ -term magnitudes is, by itself, dimensionally uninterpretable: an order-of-magnitude band (10^{-47} – 10^{-60}) spanning thirteen decades is wide, the units are not shown to match, and “a fluid solution is an exact solution of Einstein’s equations” is true of *general relativity*, not a license to import its constants into a gasifier. The contribution is to keep the *useful* part (a normalized scale-free deviation that tracks yield) and discard the *unsupported* part (physical identity).

Anti-patterns it forbids. Calling the alignment “not coincidental”; presenting the EFE as a derivation rather than a motivating analogy; conflating the *spread* of ξ collapsing toward zero with the *absolute* gain leaving its physical band.

Verification obligation. A dimensional-consistency argument showing what physical quantity k carries the units of, and a demonstration that the normalization adds predictive content beyond a plain cohort z-score of $\log k$. If a z-score does the same work, the GR framing is ornament and should be cut.

2.3.3 Contribution III — The perfect-fluid convergence index as a falsifiable conjecture

Proposal. Define $\varepsilon \equiv \text{median}|\xi| + \alpha \text{Var}(\text{sct}) + \beta \mathbf{1}[S=1]$, $\alpha, \beta > 0$, and conjecture that as operating fields smooth, $\varepsilon \rightarrow 0$, mass-closure residuals fall (superlinearly, per prediction P1), and the system approaches a perfect-fluid limit. The corpus correctly labels this a conjecture.

Why the ordering of claims matters. This is the framework’s most interesting *physical* hypothesis and also its least discharged. Its value is that it is **falsifiable**: it makes a directional, re-binnable, ablatable prediction. The contribution is to state it as a target for refutation, not a found law.

Verification obligation. The held-out test the corpus itself proposes — cross-process portability to fast pyrolysis with Λ_G recalibrated (P3), and the ablation falsifier (P4) — must be executed. A conjecture that has survived no falsification attempt is a hypothesis, not a result.

2.3.4 Contribution IV — A claim-ladder as the integrating artifact

Proposal. The contributions compose into a single release gate for any assertion the program wishes to publish. An assertion may be promoted up the ladder only if its rung’s obligation is discharged: (1) numerics — AGM/realization stable and reproducible; (2) closure — mass balance verified against a first-principles solver on held-out regimes, not only reproduced in-sample; (3) bin-level correlate — (ξ, S) monotonicity stable under re-binning and resampling; (4) orthogonality — $\xi \perp S$ demonstrated, not assumed; (5) individual-level — only with the order-of-magnitude larger, pre-registered dataset; (6) cross-domain — only after a *de novo* fit on a non-energy system reproduces the bin-level ties; (7) physical identity — only with a dimensional derivation, never on numerical proximity alone. The rungs are conjunctive *upward*: a claim at rung k that leans on an undischarged rung $j < k$ is, by construction, unsupported.

This ladder is the synthesis’s claim to systematicity. It is what converts a collection of evocative correspondences into an argument that the framework’s reach is *bounded by evidence rather than by enthusiasm* — and it is the concrete form of the “vetting procedure” the project set out to design.

2.4 The cross-domain bridge: governing a transferred claim

The biological-bridge analysis is where the program’s ambition and its evidentiary risk are both highest, so it deserves its own governance.

The legitimate core (established pattern, not established result). It is *structurally* true that any system admitting an $H(s)$ representation has a gain and a degree-balance, hence a ξ and an S ; in that narrow sense the invariants are domain-independent. It is also true that biology presents the right *shape* of problem — continuous responses (metabolic flux) alongside discrete state transitions (pathway on/off) — and that conservation-by-structure and cheap inference would be genuinely valuable there. This much is a defensible *motivation*.

The overclaim to quarantine. “The Bridge is Robust,” the 10^5 – $10^6\times$ speedup, and “the same objects appear at every scale” are not supported by any biological data in the corpus — there is no $H_{\text{bio}}(s)$ fitted to any real multi-omic series, no ξ_{bio} measured against flux, no S_{bio} validated against a cell-fate readout. The speedup compares an estimate against an unbenchmarked ODE strawman. The honest status is: **a research proposal with a clear first experiment, not a demonstrated transfer.**

The governance overlay. Borrowing the corpus’s own discipline: a cross-domain claim must be recorded as a *challengeable proposal* and may not be absorbed into the framework’s canon of results until a de-novo biological fit reproduces, at the bin level, the same orthogonal (ξ, S) alignment that gasification shows. The first experiment is well specified — insulin signaling: build $H_{\text{bio}}(s)$ from dose/time series, extract $(\xi_{\text{bio}}, S_{\text{bio}})$, test monotonicity against flux and pathway state across doses — and that, executed, would discharge rung 6.

A genuine open risk: the proxy degree of freedom. The corpus lists four admissible choices for S_{bio} (pole-zero parity, Hill cooperativity, saturation threshold, multi-state topology). This flexibility is double-edged: the very freedom that lets the framework fit any domain also lets it fit *noise*, because a proxy chosen after seeing the data is not a prediction. Treating proxy

selection as a pre-registered design decision — not a tuning knob — is a first-class obligation, not an afterthought.

2.5 Positioning crosswalk and the methodological argument

The payoff of the claim-ladder is a methodological posture that a single striking correspondence cannot earn.

Table 2.1: Demands on a surrogate framework

Demand	What it requires	Where this synthesis answers it
Conservation respect	No mass/energy violations by construction	Proper-rational $H(s)$ structure (Ch. 2.2); closure observed (2.4)
Computational tractability	Fast inference vs. PDE/ODE baselines	AGM quadratic convergence (2.1); speedup must be benchmarked, not asserted (Ch. 4)
Interpretability	Outputs map to physical meaning	Explicit poles/zeros/gain; ξ , S with stated meaning (2.3)
Honest validation	Claims scoped to evidence	Bin-level discipline (2.5); claim-ladder (3.4)
Generalization	Transfer beyond the training domain	Structural pattern real; <i>empirical</i> transfer undischarged (Ch. 4)

The decisive claim. The argument the ladder licenses is narrow but strong: *the framework is a legitimate, conservation-respecting, interpretable semi-analytic surrogate validated at the bin level on gasification, and a credible — but not yet demonstrated — template for other conservation-law domains.* This converts the program’s value from a contested statement about physics (“we found Λ in a gasifier”) into a defensible statement about method (“we built a cheap, structure-respecting surrogate and an explicit procedure for not over-claiming it”). The latter is publishable and survives interrogation; the former is not and does not.

2.6 Critical appraisal, limitations, and what must be built

A synthesis is only honest if it states where it could fail.

- **Closure is reproduced, not yet cross-checked.** “Mass closure without PDE solves” is observed *in-sample*. The owed test is a comparison against a first-principles solver on *held-out* regimes; until then, closure could be partly an artifact of fitting on the same cases. SR3 itself invites this cross-check — it should be run.
- **Two correlates from one realization are not two independent witnesses.** ξ and S are both read off the same $H(s)$. The “non-trivial, two-observable validation” claim is only as strong as a demonstrated *orthogonality* between them; absent that statistic, it risks being one witness counted twice. This is the single most important undischarged obligation for the energy-systems results.
- **The cosmological framing is a liability on a load-bearing path.** It adds rhetorical reach and subtracts credibility. The defensible version — an operational, domain-bounded normalization — should replace every instance of “not coincidental.” If a plain cohort z-score of $\log k$ matches the predictive content, the GR material belongs in a “motivation” footnote or is cut entirely.

- **The biological bridge is a proposal masquerading, in the early register, as a finding.** No biological datum has been touched. The defensible version states the *required* experiment (the insulin first-fit) and points to it as future work; presenting “the Bridge is Robust” as a conclusion would be a citation-grade failure of exactly the kind the program set out to avoid.
- **Unbenchmarked performance numbers must be removed or earned.** The 10^5 – $10^6 \times$ speedup is computed against a hypothetical. Either benchmark the AGM pipeline against a named, profiled ODE/PDE solver on a defined task, or strike the figure.

2.7 Conclusion

The contribution of this synthesis is not a new constant, a new physics, or a new universal law. It is an *architecture of restraint*: a constructive surrogate whose numerics are secure, whose structure forbids a class of conservation violations, and whose two cheap invariants are validated exactly as far as the data permits and no farther — terminating in a claim-ladder that lets an assertion rise only as its verification obligation is discharged. The foundations of that architecture — the AGM evaluation, the proper-rational realization, the bin-level (ξ, S) findings, and the corpus’s own calibrated epistemics — are real and reported as such. The integrating mechanisms — the orthogonal-invariant instrument, the demoted constitutive normalization, the falsifiable convergence conjecture, and the claim-ladder that binds them — are advanced as the candidate’s own proposals, with their open obligations stated plainly.

The wager of the program is that this is the right level of abstraction at which to make the framework *defensible* rather than merely *striking*: not by proving that a gasifier encodes the cosmos, but by proving that a fast, interpretable, conservation-respecting surrogate can be held to claims it can actually discharge — and that the “robust perfect” the program seeks is nothing more mystical than the fixed point of that discipline.

Appendix — Provenance ledger

Table 2.2: Provenance ledger

Claim or construct	Standing	Treatment in this synthesis
AGM evaluation of complete elliptic integral; quadratic convergence	Established	Accepted as foundation (Ch. 2.1)
Proper-rational $H(s)$ realization; conservation-by-structure	Established	Accepted as foundation (Ch. 2.2)
$\xi = \ln(\Lambda_S/\Lambda_G)$, $S = m - n $ definitions	Established	Accepted; basis of invariants
Mass closure without explicit PDE solves (in-sample)	Established (observed)	Accepted; held-out cross-check owed
ξ -yield and S -conversion bin-level monotonicity (200+ cases)	Established (bin-level)	Accepted at stated grain (Ch. 2.4)
Bin-level-only validation discipline (SR2-SR4)	Established (epistemic asset)	Accepted; promoted to spine (Ch. 2.5)
(ξ, S) as orthogonal two-observable instrument	Proposed	Contribution I; orthogonality proof owed
$\Lambda(x) = \Lambda_G e^{\xi(x)}$ as operational scale-free field	Proposed (defensible residue)	Contribution II; dimensional argument owed
Perfect-fluid convergence index $\varepsilon \rightarrow 0$	Conjecture	Contribution III; falsification tests owed
Cross-domain transfer of (ξ, S) to biology	Proposal, not demonstrated	Quarantined; first experiment specified (Ch. 4)
Gain “not coincidentally” the cosmological constant	Overclaim (earlier register)	Downgraded to heuristic cue / cut
10^5 – $10^6 \times$ speedup over ODE baseline	Could not be verified	Removed pending a benchmarked comparison
“The Bridge is Robust”; same objects at every scale	Could not be verified	Downgraded to motivation, not a result
Quantum-information correspondence	Open gesture	Held as open stance only; no load-bearing use

A Consolidated Solution for Provably Bounded Autonomous Driving: Instantiating the Constraint-Propagation Chain with the Witness-Gate Substrate

Purpose and method

The attached dissertation synthesis sets a single goal: engineer safety as an **unbroken chain of constraint propagation**, in which the admissible-command set contracts monotonically from perception to actuation and is finally enforced by physics rather than software. It advances four proposed contributions — a *trusted-scalar fidelity gate*, a *risk-monotone actuation algebra*, a *timing-contracted analog veto*, and a *conformance checklist* — and states their open obligations plainly.

The project corpus, developed independently, supplies a concrete mechanism that was missing from that synthesis: a **physics-grounded witness substrate** built on reduced-order-model (ROM) evidence vectors, a bilinear coupling primitive, a null-band acceptance test, a dependency-discovery loop, and a candidate hardware enforcement layer. This document consolidates those findings into one solution and shows, link by link, how the substrate *instantiates* the dissertation’s chain — discharging some obligations, sharpening others, and leaving the genuinely open ones explicitly open.

The synthesis preserves the dissertation’s evidentiary discipline. Established prior art, proposed contributions, and unverifiable claims are kept separate, and the seam between *what is known* and *what is claimed* is never blurred. Several of the project’s own headline numbers (cycle-level EPU latency, per-chip cost, the AGM/cosmological-constant observation) are treated here as **notional or motivational**, never as load-bearing evidence, for exactly the reasons the dissertation gives for the “PICAPD” architecture.

3.1 The core correspondence

The dissertation’s central claim is a monotonicity property: as one moves toward the wheels and as risk rises, the admissible-command set may only contract, never expand. The project’s central mechanism is a **representational contract**: at each decision step a bilinear *witness object* must satisfy a bounded-defect invariant, or the step is not representable at the actuator interface.

These are the same idea expressed at two levels of abstraction. The dissertation specifies *what property the chain must have*; the project specifies *a computable object that enforces that property at one stage and can be composed across stages*. The consolidation is therefore a layering, not a competition:

Table 3.1: Layering of dissertation and project mechanisms

Dissertation link	Project mechanism that instantiates it	Standing of the instantiation
Trusted-scalar fidelity gate (Contribution I)	Witness object $\mu(a, b)$ with null-band N over trusted scalar invariants (trace / spectral / gain)	Proposal $\rightarrow$ concrete candidate; one-sided-bound proof still owed
Risk-monotone actuation algebra (Contribution II)	ACCEPT / PROJECT / REJECT gate + dependency-severity bands (ADAPT/REPLAN/DEGRADE/ABORT)	Proposal $\rightarrow$ partial; exactness/non-box geometry still owed
Timing-contracted analog veto (Contribution III)	Memoryless EPU / memristive-crossbar enforcement layer	Grounded principle (Guarded Swarms); specific hardware notional
Conformance checklist (Contribution IV)	Signed witness-log + certified/learnable parameter partition	Proposal $\rightarrow$ concrete artifact; conjunctive-gate discipline carried over
Learning-theory governance bridge (Ch. 4)	Dependency-discovery loop + bounded LUT-delta updates	Detector established (LLC/stagewise); governance proposed

The remainder of the document develops each row.

3.2 Link I — The fidelity gate as a computed trusted scalar

3.2.1 What the dissertation requires

Reconstruction fidelity must not be an L2 reconstruction error. It must be a single qualified scalar φ that (i) denotes a genuine safety-relevant quantity, (ii) is bounded and saturating, (iii) **overestimates** risk whenever it errs, (iv) feeds only *constraint-tightening* maps, and (v) is re-qualified continuously and escalates monotonically toward maximum risk on any conformance failure. The inversion that makes it original is that it bounds *worst-case optimistic* error and compiles that bound into a conservative bias.

3.2.2 What the project supplies

The witness construction is a direct candidate for φ . From each ROM-valid evidence vector the substrate extracts **trusted scalar invariants** — traces, spectral summaries, and gain features — and forms the coupled witness $\mu(a, b) = a \otimes b$. The acceptance test is a bounded-defect condition on the graded-commutativity residual,

$$\Delta = \mu(a, b) - \sigma \cdot \mu(b, a), \quad \text{accept iff } (\text{tr } \Delta, \|\Delta\|, \rho(\Delta)) \in N,$$

where the null-band N is calibrated from conservation structure rather than from data fit. This maps cleanly onto the five-part contract.

3.2.3 Obligations carried forward

Two obligations transfer intact and must be discharged for any specific ODD:

1. **A one-sided-bound proof.** For a given safe set, show that the chosen invariants of μ form a valid lower bound on the true barrier margin under stated assumptions.
2. **A numerical-format argument.** The trace/spectral summaries must be computed in a format (Posit-with-quire, or a fixed-point scheme with proven one-sided rounding) whose error cannot round in the optimistic direction.

3.3 Link II — The risk-monotone actuation algebra

3.3.1 What the dissertation requires

Immediately before actuation, define admissible-command sets indexed by a normalized risk band with the **antitone** property (rising risk only shrinks the set). Each set is a closed convex axis-aligned box so that projection is firmly non-expansive (1-Lipschitz). The enforcer applies a two-stage clamp — magnitude then **rate last** — so a risk spike yields bounded jerk, not a step-lurch. Multiple envelopes compose by **intersection**, and band switching uses **asymmetric hysteresis** (tighten instantly, loosen only after dwell).

3.3.2 What the project supplies

The gate already exposes the three operators the algebra needs: **ACCEPT** / **REJECT** / **PROJECT**. The corpus explicitly notes that a reject-only gate deadlocks under uncertainty and that the third mode must project onto the nearest safe action — which is exactly the non-expansive projection the dissertation specifies. The dependency-severity ladder — **ADAPT** → **REPLAN** → **DEGRADE** → **ABORT** — is a concrete realization of antitone risk bands: each step strictly reduces actuation authority and never restores it without re-qualification. The substrate’s prohibition on averaging and union, and its use of the **conservative meet** (**intersection**) when multiple ROM classes apply, matches the composition rule one-for-one.

3.3.3 Obligations carried forward

This link inherits the dissertation’s hardest unproven theorem, and the project does not yet resolve it:

- **The exactness theorem beyond box constraints.** Sequential magnitude-then-rate clamping equals true Euclidean projection only for axis-aligned convex boxes. Real actuator limits — coupled steering/acceleration, the friction circle — are not boxes. The witness formulation may actually *help* here: because admissibility is defined by invariants of a coupling operator rather than by per-axis limits, the admissible set is naturally a (possibly non-box) sub-level set of a defect functional. But that means the project must prove either (a) that projection onto the witness-admissible set is still non-expansive and rate-bounded, or (b) that a box inner-approximation of that set is itself conservative. This is open work, not a solved problem.
- **Liveness.** A gate that always reports maximum risk satisfies every safety clause and parks the vehicle forever. The project’s own “null-band calibration robustness” and “false-reject bound” experiments are the right instruments, but the **feasibility/worst-case-bound frontier** — proving the chain permits ordinary driving — remains the central empirical question, exactly as the dissertation states.

3.4 Link III — The timing-contracted analog veto

3.4.1 What the dissertation requires

An independent analog layer enforces the actuation algebra under an explicit timing contract: intervention latency $\ll$ sensor-consistency window $\ll$ digital decision window. The guard path is hardwired comparators and latches with no programmable logic, so a hallucinating software stack cannot reach the actuator before the guard acts. On corrupted data, missed deadline, or tampered signal, the guard forces the minimal-risk envelope (maximal safe braking, zero positive acceleration) — **fail-closed**, not last-good-command coasting. The dissertation insists the specific ISA (“PICAPD”) be read as a *requirements sketch*, never as a built device.

3.4.2 What the project supplies

This is the link where the project’s hardware concept and the dissertation’s grounded principle meet — and where the same discipline must apply. The genuinely load-bearing idea, in both bodies of work, is the **Guarded Swarms analog-veto principle**: independent hardwired guard channels that can clamp an actuator within microseconds beneath any software fault. The project’s witness primitive is unusually well matched to a guard substrate because the acceptance test reduces to **analog linear algebra** (an outer-product / vector-matrix operation plus a scalar comparison), which memristive crossbars compute natively and compactly at the edge. The “memoryless” EPU framing is, at its defensible core, a statement that the guard path holds no rewritable state that a compromised stack could corrupt — which is precisely the dissertation’s privilege-isolation requirement.

3.4.3 Obligations carried forward

The project’s specific hardware claims must be downgraded to **notional requirements**, on the same grounds the dissertation applies to PICAPD:

- Cycle-level latencies (e.g., single-digit-nanosecond decisions), per-die cost figures, core counts, and register budgets are **design conjectures**, not measured properties of an existing device. Presenting them as built would be a citation failure.
- The defensible version states the *required* properties — privilege isolation, deterministic bounded latency, jitter-bounded execution, architectural rollback to a known-safe configuration on a violated safety bit, fail-closed default — and names **FPGA-accelerated kernels and lockstep RISC-V** as the credible near-term substrate for the reflexive layer, with mixed-signal “summary-and-compare” logic for the witness test.
- Analog enforcement raises a specific hazard the dissertation’s numerical-format caution anticipates: device noise and drift must be characterized and biased conservatively. Safety-critical thresholds should remain in the digital, certified partition; the crossbar accelerates μ but does not *store* the safety decision boundary unless its drift is bounded and periodically recalibrated.

3.5 Link IV — The conformance checklist as integrating artifact

3.5.1 What the dissertation requires

A single fail-closed release gate: a reconstruction may drive actuation only if, simultaneously, φ is a qualified trusted scalar; every fused input passed a pre-fusion admissibility test and carries provenance; tracking error is within its recurrent bound; the command lies in (or is non-expansively projected into) the intersection of active envelopes, rate-clamp last; no averaging or union appears on any safety path; the producing computation respected its compute/latency budget; no online update silently revised ground truth; and any failed check has already placed the system in the minimal-risk envelope. The items are **conjunctive** — the absence of a discharged guarantee is itself a fault.

3.5.2 What the project supplies

The project’s **signed witness-log** is the concrete realization of this checklist as a runtime artifact. Each decision emits a structured record — timestamp, ROM-class and pathway identifiers, evidence-vector hashes, defect summaries $S(\Delta)$, gate outcome (accept/reject/project), discovered dependencies, and the handler action taken. This record *is* the per-step proof token the dissertation’s gate demands, and it is exactly what converts “we tested hard” into “every command

carries a discharged guarantee.” The project’s **certified-vs-learnable parameter partition** (immutable certified thresholds; bounded-adaptable tolerances within a certified envelope; free-learning history) operationalizes the checklist’s separation between what may change online and what may not — and the cryptographic signing of update records provides the provenance and version/rollback discipline the conjunctive gate requires.

The match is strong enough that the dissertation’s checklist can be adopted essentially verbatim as the project’s **release gate specification**, with each clause bound to a concrete witness-log field.

3.6 The learning-theory bridge — governing an updating policy

3.6.1 The shared problem

Both bodies of work confront a policy that updates online and may *quietly* acquire a new behavior. The dissertation’s honest position is that the analog veto is the real-time guarantee and learning governance is a slower assurance layer; conflating their timescales would overstate the result.

3.6.2 The consolidated answer

The project’s architecture already enforces this timescale separation, which is a point of agreement worth stating as a strength rather than a caveat:

- **Detector (established).** The dissertation supplies a real, online-estimable signal: a developmental phase transition appears as a coupled regret-drop / Local-Learning-Coefficient jump, with the singular-MDL result bounding how far lossy perception can compress before destroying causally necessary structure. The project’s dependency-discovery loop supplies a complementary fast signal: a **spike in witness rejects clustered by ROM class** indicates that a latent variable (road grade, payload, surface friction, actuator degradation, hidden agent intent) is missing from the conditioning. Formally, *a dependency is any latent variable whose omission pushes the witness-defect statistics out of the null band under some candidate action family.*
- **Governance (proposed).** The project’s bounded-update rule — the dependency module emits **LUT deltas within a certified envelope, never raw weights**, and logs/signs every change — is a concrete implementation of the dissertation’s anti-silent-drift principle: any update that alters the policy’s effective judgments is recorded as a challengeable event rather than absorbed into the canon. This should be presented, as the dissertation insists, as an architectural pattern (proposer / verifier / adjudicator separation), not as an established governance theory.
- **The implicit-bias caution (genuine open risk).** The dissertation’s sharpest learning-theory warning carries directly into the project: the optimizer’s implicit geometric bias, which makes overparameterized policies generalize, can *degrade* robustness when it is misaligned with the threat geometry. A witness-gated policy that generalizes beautifully on clean data may still be systematically fragile in the exact direction a rare scene or adversary exploits — and the gate only catches that fragility if the chosen invariants span that direction. Alignment between optimizer bias and the safety threat geometry must be a first-class design constraint on the choice of invariants, not an afterthought.

The honest claim is therefore unchanged: the analog veto bounds the real-time consequence of a hallucination; the dependency-discovery / LLC layer can only *quarantine a capability* between drives. The project’s design respects this boundary.

3.7 Standards crosswalk

The consolidated solution inherits the dissertation’s certification posture, with the project’s mechanisms attached to each demand.

Table 3.2: Standards crosswalk

Standard	What it demands	Where the consolidated solution answers it
ISO 26262 (ASIL)	Mitigate E/E faults; freedom from interference	Hardwired analog veto with deterministic bounded latency; lockstep RISC-V substrate; certified/immutable threshold partition; conservative numerical format
ISO 21448 / SOTIF; UL 4600	Bound the “unknown unsafe” with no hardware fault	Trusted-scalar witness gate + risk-monotone projection make unsafe commands non-representable at the actuator interface
IEEE 2846 / RSS / SFF	Minimum reasonable assumptions; behavior across ODD transitions	Rate-clamp-last bounded-jerk entry into stricter envelopes; asymmetric hysteresis; dependency-severity ladder
AEC-Q100 / ISO 16750-3	Thermomechanical and vibration survivability	Vibration/noise floor sets the real lower bound on any witness threshold; analog-device drift bounded and recalibrated

The **decisive claim** is the dissertation’s, now mechanized: even under severe algorithmic hallucination in perception or planning, the actuation-boundary witness gate forces the vehicle into a provable Minimum Risk Condition. This converts SOTIF compliance from a statistical claim about how rarely the stack errs into a structural claim about what an error *can do* — and it does so without violent braking at ODD transitions, because the same rate-clamp-last discipline that prevents lurching carries the vehicle smoothly into the more conservative RSS parameters that fog or rain demand.

3.8 Critical appraisal — what is solved, sharpened, and still open

Honesty about standing is the program, not a footnote to it.

What the project genuinely contributes to the goal.

- A *computable* trusted scalar (the witness μ and its null-band) where the dissertation had only a contract.
- A *runtime artifact* (the signed witness-log) that turns the conjunctive checklist from a specification into an auditable proof stream.
- A *learnable-but-bounded* adaptation mechanism (dependency discovery emitting certified-envelope LUT deltas) that operationalizes anti-silent-drift governance.
- A *substrate match*: the witness test is analog linear algebra, which is well suited to a hardwired microsecond guard path — the right shape for the analog veto.

What the consolidation sharpens but does not close.

- *The one-sided-bound proof*. The witness must be shown to err only toward “reject.” This is the single most important open obligation; until it is discharged for a given safe set, the fidelity gate is a candidate, not a guarantee.

- *The exactness theorem off the box.* Projection onto the witness-admissible set must be proven non-expansive and rate-bounded for realistic (non-box) actuator geometry, or shown conservative under a box inner-approximation.
- *Liveness / feasibility.* The trade-off frontier between worst-case bound and ordinary drivability is the central empirical question and is open.

What must not masquerade as built.

- The EPU’s specific latency, cost, core-count, and register figures are notional, exactly as the dissertation treats PICAPD. The defensible posture states *required* properties (privilege isolation, bounded rollback latency, jitter-bounded fail-closed execution) and points to FPGA / lockstep-RISC-V as the realizable substrate.
- The motivational origin story (AGM elliptic kernels; the observation that gain parameters fall near cosmological-constant magnitudes) is intellectual provenance and a source of useful physics-kernel intuition — it is **not** evidence on any safety-bearing path and should never appear in a certification argument.

3.9 A staged program to discharge the obligations

The consolidation implies a concrete order of work, each stage retiring a named risk.

1. **Witness one-sidedness (theory + numerics).** Prove, for one well-characterized ODD and one safe set, that the chosen invariants of μ lower-bound the true barrier margin, and specify a numerical format whose rounding cannot be optimistic. *Retires the Link-I obligation.*
2. **Separability and false-reject bounds (emulator).** Show on an FPGA/mixed-signal prototype that safe and unsafe trajectories produce statistically separable $S(\Delta)$ signatures across ROM classes, with bounded false-reject rates under device noise. *Begins to address liveness.*
3. **Projection geometry (theory).** Prove non-expansiveness and rate-boundedness of projection onto the witness-admissible set for coupled steering/acceleration limits, or establish a conservative box inner-approximation. *Retires the Link-II exactness obligation.*
4. **Microsecond fail-closed guard (hardware).** Demonstrate a hardwired guard path that clamps to the minimal-risk envelope within the timing contract, with no rewritable state in the guard. *Validates Link III as principle; keeps hardware claims honest.*
5. **Policy-drift immunity (integration).** Repeatedly mutate the foundation policy and show invariants are never bypassed; show the dependency loop quarantines a capability between drives without relaxing enforcement. *Validates Link IV and the governance bridge.*
6. **Cross-domain transfer (generalization).** Re-run the same gate on a second platform (manipulator or mobile robot) to substantiate that the contribution lives at the level of invariants and representability, not vehicle-specific tuning.

3.10 Conclusion

The dissertation’s contribution is an *architecture of refusal*: a chain in which each stage can only narrow what the next may do, terminating in an analog layer that enforces the narrowing with electricity rather than code. The project corpus supplies the missing operational core

of that architecture — a physics-grounded witness object whose bounded-defect invariant is the computable form of the trusted scalar, whose accept/reject/project gate is the actuation algebra, whose signed log is the conformance checklist, and whose dependency-discovery loop is the anti-silent-drift governance — together with a substrate (memristive/FPGA enforcement) whose native operation is exactly the witness test.

Consolidated, the two yield one solution and one wager: that autonomous driving becomes *certifiable* rather than merely *tested* not by proving the stack never errs, but by proving — through a one-sided witness, a non-expansive projection, and a hardwired fail-closed veto — that when it errs, the error has nowhere unsafe to go. The established components are cited as such; the integrating mechanisms are advanced as the program’s own proposals; and the three obligations that gate the whole claim — witness one-sidedness, projection exactness off the box, and the liveness frontier — are stated plainly as the work that remains.

Appendix — Provenance ledger (consolidated)

Table 3.3: Provenance ledger

Claim or construct	Standing	Treatment in this solution
CBF / RoM–FoM layered control / RTF / Poisson safety / neural & gameplay filters	Established	Accepted as foundation
Sensing & numerics; Posit vs IEEE-754; symplectic integration; AEC-Q100	Established	Accepted as substrate; numerical-format constraint binds the witness
SLT / LLC / stagewise transitions; implicit-bias caution	Established	LLC accepted as slow detector; implicit-bias alignment made a design constraint
ISO 26262 / SOTIF / UL 4600 / RSS / SFF / IEEE 2846	Established	Accepted; certification cross-walk (§7)
Guarded Swarms analog-veto principle	Established (verified)	Accepted; basis of Link III
Witness object $\mu(a, b)$ + null-band N over trusted scalar invariants	Project proposal	Instantiates the trusted scalar (Link I); one-sided-bound proof owed
ACCEPT/PROJECT/REJECT gate + dependency-severity bands	Project proposal	Instantiates the actuation algebra (Link II); exactness off the box owed
Signed witness-log + certified/learnable partition	Project proposal	Instantiates the conformance checklist (Link IV)
Dependency-discovery loop + bounded LUT-delta updates	Project proposal	Instantiates anti-silent-drift governance (§6)
EPU / “Queen Bee” hardware specifics (latency, cost, core/register counts)	Could not be verified as built	Treated as notional requirements sketch; FPGA / lockstep-RISC-V named as substrate
AGM elliptic-kernel origin; cosmological-constant magnitude observation	Motivational / unverified for safety	Intellectual provenance only; never on a safety-bearing path

Mathematical Consolidation of the Constraint-Propagation Program: From Harmonic-Analytic Foundations to a Provably Bounded Actuation Boundary

Purpose and reading instructions

The goal synthesis advances one thesis: safety should be engineered as an **unbroken chain of constraint propagation**, in which the admissible-command set contracts monotonically from perception toward actuation and is finally enforced by physics rather than software. That synthesis is candid that several of its integrating mechanisms — the trusted-scalar fidelity gate, the risk-monotone actuation algebra, the timing-contracted veto — are *proposed* contributions carrying explicit verification obligations rather than settled prior art.

This document does something the goal synthesis could not do from its own corpus: it supplies the **mathematical substrate** for those obligations. The eight project findings are not decoration for the architecture; they are the analysis on which two of its load-bearing claims either stand or fall. The contribution here is therefore a *mapping with discharge*: for each link in the chain we name the established mathematics that grounds it, we prove what can be proved, and we mark the seam where mathematics runs out and engineering conjecture resumes. The discipline borrowed from the goal synthesis is preserved without exception — the boundary between *what is proven* and *what is owed* is never blurred.

The single most consequential result below is that the actuation algebra’s owed “exactness theorem” is **partially discharged** by elementary convex analysis already present in the corpus, and that the precise location where it *fails* can now be stated exactly, with the failure mode named in the language of rank-deficient projection.

4.1 The consolidated toolkit

The project findings reduce to six reusable instruments. Each is established mathematics; none is original here.

- **(T1) Separability of Cartesian-product operators (Finding 1, Finding 7).** A separable two-dimensional transform factors into independent one-dimensional passes, $F = A_M f A_N^\top$, requiring only that the operator on each axis match that axis’s dimension. For an orthogonal transform the inverse is the transpose, $A^{-1} = A^\top$, so reuse is cheap and no inversion is incurred. The dimensional discipline underneath this — that $\mathbb{F}^n$ has *exactly* n coordinates, and that lower-dimensional data enters higher-dimensional spaces only by explicit zero-padding $\mathbb{F}^k \hookrightarrow \mathbb{F}^n$ — is the same discipline that governs every state-lift and command space below.

- **(T2) Fast forward models for wave-like perception (Finding 2, Finding 6).** The naive Fourier-integral-operator (FIO) evaluation over paired $N \times N$ spatial and frequency grids costs $O(N^4)$; angular-wedge partition, phase linearization with low-rank separation, and nonuniform FFTs reduce this to $\approx O(N^{2.5} \log N)$. FIOs are the correct forward model for wave propagation and therefore for radar/LiDAR-style imaging; microlocal analysis tracks the geometry of singularities (edges, obstacles) carried by such operators.
- **(T3) Exact frequency-domain filtering (Finding 3).** Linear convolution equals N -point circular convolution **only after** zero-padding both operands to a common length $N \geq P + Q - 1$; otherwise wrap-around contaminates the output. The two-dimensional rule pads each axis to at least $H + K - 1$ and $W + L - 1$.
- **(T4) Well-posedness of the continuous limit (Finding 4, Finding 5).** On the tempered space, $\mathcal{F} : \mathcal{S}' \rightarrow \mathcal{S}'$ is a topological isomorphism and convolution maps to pointwise multiplication continuously; regularization by a mollifier converges, $T * \rho_\varepsilon \rightarrow T$ in $\mathcal{D}'$. The ambient topological vector spaces are controlled by countable seminorm families and are **complete** (Fréchet / LF structure), which is exactly the property that guarantees the limits, fixed points, and convergent estimators invoked downstream actually exist.
- **(T5) Resolution floors (Finding 6).** Uncertainty principles impose an irreducible lower bound on simultaneous time–frequency localization — a “what you cannot do” that mirrors, in the analytic domain, the MEMS noise floor the goal synthesis treats as the hard lower bound on any real safety threshold.
- **(T6) Projection onto convex constraint sets (Finding 8).** A rank-1 operator $A = uv^\top$ splits any vector uniquely into an along-image component and an in-kernel component, is invertible only on its one-dimensional image, and admits a Moore–Penrose pseudoinverse implementing the canonical isomorphism between coimage and image. This is the elementary instance of the general fact used decisively below: **projection onto a closed convex set is firmly non-expansive (1-Lipschitz), and the decomposition into “feasible component plus residual” is unique.**

4.2 The mapping: each link and its mathematical ground

The chain runs perception $\rightarrow$ fusion $\rightarrow$ fidelity scalar $\rightarrow$ control lift $\rightarrow$ actuation algebra $\rightarrow$ analog veto. We attach the substrate link by link, then prove the two links that the mathematics genuinely settles.

Two rows carry real proof obligations that the corpus can now address. We take them in order of how far the mathematics reaches: the actuation algebra (mostly dischargeable) and the fidelity scalar (only its prerequisites dischargeable).

4.3 Discharging the actuation algebra’s exactness theorem (Contribution II)

The goal synthesis owes a theorem: that the proposed **sequential magnitude-then-rate clamp equals the true Euclidean projection** onto the admissible set, with the enforcer guaranteed never to amplify an oscillation. We prove the parts the mathematics settles, then isolate precisely the part it does not.

4.3.1 The feasible case is exact and is separable

Let the pre-actuation command live in a fixed space $\mathbb{R}^n$ (T1’s dimensional discipline: exactly n actuator channels, no implicit length change). An axis-aligned magnitude box is a Cartesian

Table 4.1: Mapping each link to mathematical ground

Chain link	Grounding	What the mathematics contributes	Standing
Real-time perception transforms	T1, T2	Separable passes + fast-FIO complexity make the forward model fit the compute/latency budget	Grounded
Frequency-domain filtering in the loop	T3	Padding to $N \geq P + Q - 1$ removes wrap-around; a prerequisite for the fidelity scalar’s honesty	Grounded prereq.
Continuous well-posedness / stability	T4	$\mathcal{S}'$ makes the discrete operations stable discretizations of a topological automorphism	Grounded
Trusted-scalar fidelity gate (I)	T3, T5, num.	Sets the honest floor and the no-optimistic-rounding prerequisite; not the one-sided bound itself	Prereqs grounded; core owed
Reduced-order $\rightarrow$ full-order lift	T1, T6, dim.	The RoM state embeds into the FoM space ($\mathbb{F}^k \hookrightarrow \mathbb{F}^n$); tracking is a statement about that embedding	Framed; recurrence accepted
Risk-monotone actuation algebra (II)	T6, T1, dim.	Box projection is separable and non-expansive — discharges the feasible case of the owed theorem	Partially proven (§3)
Timing-contracted analog veto (III)	—	Hardware/timing argument; outside the analytic substrate	Accepted (Guarded Swarms)

product of intervals,

$$M = \prod_{i=1}^n [a_i, b_i].$$

Proposition 4.1 (separability of box projection). *The Euclidean projection onto M acts coordinatewise:*

$$[\Pi_M(x)]_i = \text{clip}(x_i, a_i, b_i) = \min(\max(x_i, a_i), b_i).$$

Proof. Minimizing $\sum_i (x_i - y_i)^2$ subject to $a_i \leq y_i \leq b_i$ has an objective that **separates** over coordinates; each coordinate minimizes independently, and the constrained minimizer of a one-dimensional quadratic on an interval is the clip. $\square$

This is the convex-analytic mirror of T1: a constraint set that is a Cartesian product projects as a *product of projections*, exactly as a separable operator factors into *independent axis passes*. Separability is the same structural fact appearing in two registers.

Proposition 4.2 (non-expansion). *Projection onto any closed convex set is firmly non-expansive: $|\Pi_C(x) - \Pi_C(y)| \leq |x - y|$. In particular the scalar clip is 1-Lipschitz, and the product map inherits the bound.*

Proof sketch. Firm non-expansiveness of the metric projection onto a closed convex subset of a Hilbert space is standard; the coordinatewise clip is the projection onto an interval, hence 1-Lipschitz, and a Cartesian product of 1-Lipschitz maps is 1-Lipschitz in the Euclidean norm. $\square$

Proposition 4.2 discharges in full the goal synthesis’s claim that “projecting any proposed command onto the box is firmly non-expansive, guaranteeing the enforcer can never amplify an oscillation injected by corrupted software.” The guarantee is not a hope about a particular implementation; it is a property of metric projection onto convexity, and it survives any later generalization of the constraint set so long as that set remains closed and convex (this matters in §3.3).

Proposition 4.3 (the rate clamp is also a box, and the intersection is a box). *The rate limit constrains the increment, $u_t - u_{t-1} \in \prod_i [-r_i, r_i]$, which is an axis-aligned box $R(u_{t-1})$ in command space centered at the previous command. The intersection of two axis-aligned boxes is the componentwise intersection of intervals, hence again an axis-aligned box,*

$$M \cap R(u_{t-1}) = \prod_i [\max(a_i, u_{t-1,i} - r_i), \min(b_i, u_{t-1,i} + r_i)],$$

whenever this is nonempty.

Corollary 4.1 (exactness, feasible case). *When $M \cap R(u_{t-1}) \neq \emptyset$, sequential clipping — clip onto M , then clip onto R — equals the single projection onto $M \cap R$, and equals it independent of the clamp order. Per coordinate, composing two interval-clips with overlapping intervals yields the clip onto their intersection (monotonicity and idempotence of the clip make the composition land at the projection’s fixed point). Combined with Proposition 4.1, this is the **exactness theorem the goal synthesis owed, proven for the feasible regime**. The effective envelope under multiple constraints is their intersection (a meet), never a union or average — which is precisely the admissible algebra Contribution II demands.*

4.3.2 Where exactness genuinely fails — and what the failure is

The owed theorem does **not** hold globally, and the corpus now lets us say exactly why. The substantive content of “rate-clamp last” is not which set the command lands in (in the feasible case both orders land in the intersection). It is the behavior when a **risk spike shrinks M so far that $M \cap R(u_{t-1}) = \emptyset$** — the new magnitude target is unreachable in one rate-limited step.

In that infeasible regime there is no projection onto the intersection, because the intersection is empty. The clamp ordering is then a **prioritization rule**, not a projection: rate-clamp-last sacrifices the (unreachable) magnitude target to preserve the bounded-jerk rate envelope, ratcheting the command toward the tighter box at maximal admissible jerk — the goal synthesis’s “graceful distance-decay” rather than a step-lurch. Magnitude-last would instead honor the magnitude box and violate the rate envelope, producing the unbounded step the architecture is built to forbid.

This is exactly the structure of Finding 8 (T6). When the full constraint cannot be met, the command decomposes uniquely into a component along the satisfiable (rate) direction and a residual lying in the direction of the violated (magnitude) constraint; the pseudoinverse picks out the feasible component and the residual is carried as bounded, decaying error. The honest theorem statement is therefore:

Sequential magnitude-then-rate clamping equals Euclidean projection onto the admissible box **whenever the rate-feasible and magnitude-feasible sets intersect**; when they do not, rate-last clamping is the **lexicographic (rate-prioritized) projection**, which is the correct safety rule precisely because it bounds jerk at the cost of a transient, monotonically decaying magnitude residual.

The first half is now proven (§3.1). The second half is **defensible but not yet a closed theorem**: it requires showing the residual decays monotonically to zero once the intersection becomes nonempty again, under the actuator’s realizable dynamics. That decay claim is the genuine open obligation, and it is a control-theoretic statement layered on top of the projection algebra, not a contradiction of it.

4.3.3 The non-box obstruction (coupled limits)

The goal synthesis flags that real actuator limits — a friction circle $\sqrt{a_x^2 + a_y^2} \leq \mu g$, coupled steering/acceleration envelopes — are **not** axis-aligned boxes. The consequence is precise and bounded:

- **Separability fails.** A Euclidean ball is not a Cartesian product, so Proposition 4.1 does not apply and **per-coordinate clamping is no longer the projection**. Sequential clamping can land outside the true admissible set or short of it.
- **Non-expansion survives.** The ball is closed and convex, so by Proposition 4.2 its metric projection (radial scaling toward the boundary) remains firmly non-expansive. The *safety* property — no amplification of injected oscillation — is therefore preserved exactly; only the cheap separable *implementation* is lost.

The discharge is thus partial in a clean way: the **guarantee** generalizes (any convex envelope projects non-expansively), the **implementation** does not (clamping must be replaced by true convex projection, in closed form for a ball, or by an alternating-projection iteration for box-meet-ball). Restricting Contribution II’s claims to the box regime, or replacing the clamp with a genuine convex projection, both close the gap; asserting separable clamping for non-box geometry would not.

4.4 Grounding the fidelity-scalar prerequisites (Contribution I)

The trusted scalar φ must be a one-sided bound: it may err only by **overestimating** risk. The corpus does not supply that bound — constructing it for a given safe set is the dissertation’s central open problem — but it discharges two necessary conditions without which the bound leaks.

- **Honesty floor (T5).** The scalar is required to be bounded and saturating and to denote a genuine safety-relevant margin. The uncertainty principle sets the analytic floor on resolvable detail, paralleling the MEMS noise floor of the substrate: φ cannot honestly claim a margin finer than the joint time–frequency (or sensor-noise) floor permits. Any contract clause asserting sub-floor precision is vacuous, and the consolidated bound on resolution makes that explicit.
- **No-optimistic-leak prerequisite (T3, T4, numerics).** If frequency-domain filtering in the perception path is computed without padding to $N \geq P + Q - 1$, wrap-around injects energy that can make a reconstruction look *cleaner* than it is — error in the optimistic direction, exactly the direction the contract forbids. Finding 3 is therefore not merely an efficiency note; it is a **correctness precondition** for the trusted scalar. The tempered-distribution setting (T4) supplies the matching guarantee in the continuous limit: because $\mathcal{F}$ is a topological automorphism on $\mathcal{S}'$ and convolution maps continuously to multiplication, the discrete filter is a stable discretization rather than a source of uncontrolled drift — provided, per the substrate’s numerical-format argument, the arithmetic rounds on the safe side.

What remains owed is unchanged from the goal synthesis: a proof that the chosen φ is a valid one-sided barrier-margin witness, plus the **liveness** argument that the gate is not so conservative it forbids ordinary driving. The mathematics here removes two ways the bound could silently fail; it does not manufacture the bound.

4.5 Consolidated solution statement

Reassembled, the chain reads as follows, with each clause now annotated by its standing.

1. **Perception** runs as separable, fast-FIO forward models within budget (T1, T2): *grounded*.
2. **Filtering** pads to $N \geq P + Q - 1$ so no wrap-around can flatter a reconstruction (T3, T4): *grounded as the trusted scalar’s precondition*.

3. **The fidelity scalar** is held below the resolution/noise floor and computed without optimistic rounding (T5, numerics): *prerequisites grounded; the one-sided bound and its liveness remain the central open problem.*
4. **The state lift** embeds the reduced-order model into the full-order command space under fixed dimension (T1, T6, dimension discipline); the recurrent tracking bound is accepted control-theoretic foundation.
5. **The actuation algebra** projects every command onto the intersection of active envelopes. In the feasible case this projection is **separable and exact** (Prop. 4.1, 4.3) and **provably non-amplifying** (Prop. 4.2) — the owed exactness theorem, discharged. In the infeasible (risk-spike) case, rate-last clamping is the rate-prioritized projection whose residual-decay claim is the remaining obligation (§3.2). For non-box geometry the non-amplification guarantee survives but separable clamping must yield to true convex projection (§3.3).
6. **The analog veto** enforces the narrowing in hardware: accepted from verified prior art, outside the analytic substrate.

The decisive claim the goal synthesis sought — *that even under severe perception/planning hallucination the actuation boundary forces a provable Minimum Risk Condition* — now rests on a proven core. The non-amplification property at the enforcement boundary is a theorem about convex projection, not an empirical hope; the feasible-case exactness is a theorem about separability; and the two genuinely open links (the trusted scalar’s one-sided bound with liveness, and the infeasible-case residual decay) are isolated, named, and reduced to statements over well-posed, complete spaces (T4) where the needed limits and fixed points are guaranteed to exist.

Provenance ledger

Table 4.2: Mathematical consolidation provenance

Item	Source	Standing / Treatment
Separability of Cartesian-product operators; orthogonal $A^{-1} = A^\top$	Finding 1	Established; grounds Prop. 4.1
Fast FIO $O(N^4) \rightarrow O(N^{2.5} \log N)$; microlocal forward model	Findings 2, 6	Established; grounds perception compute-budget link
Linear = circular convolution after padding to $N \geq P + Q - 1$	Finding 3	Established; grounds the trusted-scalar correctness precondition
Tempered distributions; $\mathcal{F}$ automorphism; mollifier convergence	Finding 4	Established; grounds continuous well-posedness / stability
Completeness; countable-seminorm / Fréchet-LF structure	Finding 5	Established; guarantees existence of limits, fixed points, estimators
Uncertainty principle resolution floor	Finding 6	Established; grounds the trusted scalar's honesty floor
$\mathbb{F}^n$ exact dimension; zero-pad embedding $\mathbb{F}^k \hookrightarrow \mathbb{F}^n$	Finding 7	Established; dimensional discipline for command space and RoM→FoM lift
Rank-1 decomposition; non-expansive projection; pseudoinverse	Finding 8	Established; grounds Props. 4.2–4.3 and the residual decomposition
Feasible-case exactness theorem (sequential clamp = box projection)	This consolidation	Proven (§3.1); discharges Contribution II in the feasible regime
Infeasible-case rate-prioritized projection; residual monotone decay	This consolidation	Defensible, not closed; stated as remaining obligation (§3.2)
Trusted scalar one-sided bound + liveness	Goal synthesis	Open; prerequisites grounded (§4); core obligation unchanged
Analog veto / timing contract	Guarded Swarms (verified)	Established; outside analytic substrate

Consolidated Solution: The Ghost Portfolio as an Architecture of Refusal — Mapping the 12-Dimensional Technical Positioning onto a Provably Bounded Autonomous-Driving Stack

Date. May 23, 2026

Status. Integrating synthesis — companion to the four positioning documents and the dissertation synthesis *Constraint Propagation from Epistemology to Actuation*.

Method. Re-sort Ghost’s seven differentiated assets along the dissertation’s monotone-contraction chain; apply the dissertation’s evidentiary contract to each claim.

5.1 The goal, stated as an engineering target

The dissertation does not ask for a better controller, sensor, or policy. It asks for an **architecture of refusal**: a chain in which the set of actuator commands a stage may authorize can only *contract* as one moves toward the wheels and as estimated risk rises, with the final contraction enforced by physics rather than software. Formally, the admissible-command set is antitone in risk and the terminal enforcement is non-programmable.

The four positioning documents already contain most of the components such a chain requires. They simply organize them around the wrong axis — competitive differentiation rather than constraint flow. The consolidation is therefore a *re-projection*: the same assets, re-indexed by their function in the loop.

The central finding, stated up front: **Ghost’s portfolio is strong precisely where the dissertation’s chain terminates (the reflexive electrical veto and the control-theoretic narrowing above it) and weak or silent precisely where the chain begins (the epistemic gate and the numerical substrate)**. The distributed assets (Byzantine consensus, population scalability) sit *outside* the single-vehicle goal entirely and belong to a later multi-agent extension. This is a clean division of labor, not a defect — but only if the seams are drawn honestly.

5.2 The evidentiary contract, inherited from the goal

The dissertation’s load-bearing methodological move is its refusal to blur “what is known” with “what is claimed.” A consolidation that imported the positioning documents’ validated-sounding language would defeat the purpose of the goal document. So each Ghost asset below is tagged with the same three-valued standing the dissertation uses:

- **Established** — independently attested engineering or theory.
- **Proposed** — a defensible synthesis advanced as a hypothesis, with verification owed.
- **Notional / unverified** — a requirements sketch or claim presented in the corpus as built but not demonstrated.

Applied honestly, most of Ghost’s headline numbers fall into the second and third categories. That is not fatal. The dissertation’s own central contributions are *proposals*; a doctoral-grade synthesis is a defensible new arrangement of established parts. The discipline is only that the arrangement never be sold as a measured result.

5.3 Stage-by-stage consolidation

5.3.1 Stage 0 — The epistemic gate (perception): the trusted scalar φ

What the goal requires. A single qualified scalar that answers not “how accurate is my reconstruction” but “can the detail I discarded only ever *shrink* the admissible action set.” It must denote a genuine barrier-margin or causal-sufficiency witness, saturate, overestimate risk when it errs, feed only constraint-*tightening* maps, and be re-qualified continuously.

What Ghost supplies. Partially, and indirectly. The physics-informed learning work (the PDE-residual penalty $L = L_{\text{data}} + \lambda_1 L_{\text{PDE}} + \lambda_2 L_{\text{BC}}$, patents 11,983,889 and 11,971,958) restricts the hypothesis space using equation-class structure. That restriction is the *raw material* for a one-sided bound: a margin computed against a physics-constrained model has a principled error structure rather than an arbitrary one. The “compositional safety” idea (hyperbolic reachability $\partial R / \partial t = c|\nabla R|$ composed with ODE vehicle dynamics) is exactly a barrier-margin witness of the kind φ is supposed to be.

The gap. Ghost’s portfolio computes margins; it does not anywhere prove they are *one-sided*. Nothing in the four documents establishes that the reachability margin lower-bounds the true margin, nor that it is computed in a format that cannot round optimistically. Both are the dissertation’s explicit Contribution-I obligations, and both are unmet in the corpus.

Standing: physics priors *Established* (patents issued); use of the reachability margin *as* a trusted scalar *Proposed*; the one-sidedness and conservative-rounding proofs *owed*.

5.3.2 Stage 1 — The numerical substrate

What the goal requires. Any scalar that gates actuation must be computed in a format whose rounding error is bounded *on the safe side*. The dissertation treats numerical format as a safety variable — IEEE-754 single precision can drive an EKF covariance to lose positive-definiteness within hundreds of seconds; Posit-with-quire tracks a double-precision reference across the horizon.

What Ghost supplies. Nothing explicit. The positioning documents discuss compute in terms of energy and latency (< 200 pJ/op, < 5 ns) but never in terms of numerical conservatism. This is a genuine hole, and it is the cheapest one to fill: the EPU’s fixed-function constraint checkers (Stage 3) are an ideal place to mandate a saturating, safe-side-rounding numeric format because they are stateless and exhaustively analyzable.

Standing: *Gap*. This is the lowest-cost, highest-leverage addition the consolidation identifies — a specification item, not a research program.

5.3.3 Stage 2 — Control-theoretic narrowing: MPC, compositional safety, and the actuation algebra

What the goal requires. Layered reduced-order/full-order control with control-barrier functions and recurrent tracking functions; then, at the boundary before actuation, a *risk-monotone actuation algebra* — admissible sets that are closed convex axis-aligned boxes, projection onto which is firmly non-expansive (1-Lipschitz), implemented as a magnitude-then-rate clamp with the **rate clamp applied last** so a risk spike yields bounded jerk rather than a step-lurch. Composition of multiple envelopes is **intersection**, never union or average; band switching uses **asymmetric hysteresis**.

What Ghost supplies. The strongest *conceptual* alignment in the portfolio after the veto layer. Ghost’s control architecture is MPC-dominant by design and explicitly equation-appropriate — MPC for the hyperbolic/transport/parabolic domains, RL confined to ODE actuation. The dissertation’s actuation algebra is, structurally, a hard-constraint MPC projection; Ghost’s “MPC for all wave-domain constraints” is the same commitment stated competitively. The compositional safety design ($\arg \min \|u\|$ subject to hyperbolic and ODE constraints) is precisely the RoM/FoM layering the dissertation accepts from the CBF/RTF literature.

The refinement the consolidation adds. Ghost frames MPC as the primary *decision* engine. The dissertation’s discipline reassigns it: MPC is the *narrowing* layer, and the non-expansive clamp ordering (the part Ghost does not specify) is a safety property the consolidated stack must adopt. Ghost’s portfolio gives the constraint solver; the dissertation gives the algebra that constrains how the solver’s output is projected.

Standing: MPC-dominant control *Established*; compositional PDE+ODE safety *Proposed* (design phase); the clamp-ordering exactness theorem *owed* — and owed for *non-box* geometry (friction circles, coupled steering/acceleration limits), which the corpus’s axis-aligned framing does not cover.

5.3.4 Stage 3 — The reflexive electrical veto (the strongest match)

What the goal requires. An independent analog layer under an explicit timing contract — intervention latency $\ll$ sensor-consistency window $\ll$ digital decision window — built from hardwired comparators and latches with **no programmable logic in the guard path**, so a compromised or hallucinating software stack cannot reach the actuator first. It must be **fail-closed**: on corrupted data, a missed deadline, or a tampered path, it forces the minimal-risk envelope. The dissertation treats its “PICAPD ISA” as a *notional requirements sketch* — privileged non-writable safety registers, deterministic bounded latency, architectural rollback — and points to FPGA / lockstep-RISC-V as the realizable substrate.

What Ghost supplies. This is where the two corpora *converge independently*, and that convergence is the most defensible result of the consolidation. Ghost’s EPU is specified as **combinational logic only** — **no sequential state machines, fixed-function checkers, deterministic, ASIL-D by design because it has no hidden failure modes**. That is, almost line for line, the dissertation’s requirements for the guard path. The dissertation arrives at “no programmable logic, exhaustively verifiable, bounded latency” from the certification side; Ghost arrives at “combinational only, finite input space, exhaustive verification feasible” from the silicon side. Two independent derivations of the same artifact is the closest thing to validation either document offers.

The reassignment. Ghost positions the EPU as a general constraint-satisfaction engine for *every* decision. The dissertation’s discipline narrows its role: the combinational, exhaustively-verifiable EPU should be the **enforcer/veto**, not the primary planner. The part of a stack that must be provably safe is small and stateless; the part that must be smart is large and fallible. Ghost’s EPU is the right substrate for the former, and the FPGA prototype the corpus reports as validated is exactly the dissertation’s named near-term realization. The < 5 ns figure matters here not as a competitive headline but as the satisfaction of the timing contract’s first inequality.

Standing: combinational ASIL-D-by-construction design *Proposed but well-motivated* (FPGA-stage, ASIC notional); the role *as the dissertation’s analog veto* is the consolidation’s own claim; the < 5 ns ASIC figure *notional* until tape-out. Critically — and this is the dissertation’s exact warning about “PICAPD” — the EPU must not be presented as a built device. It is a requirements-satisfying design with an FPGA witness.

5.3.5 Cross-cutting A — Governing an updating policy

What the goal requires. Online detection that a learning policy has *quietly* acquired a new behavior (the established LLC / stagewise-transition signal, estimable via SGLD), feeding an **anti-silent-drift** governance overlay structured as proposer / verifier / adjudicator separation, with capabilities quarantined between drives rather than vetoed in real time.

What Ghost supplies. The *governance scaffolding* but not the *detector*. Ghost’s “cognitive continuity tags,” “behavioral attestation,” 175-step audit logs, and decision-thread IDs are a recognizable proposer/verifier/adjudicator pattern — the architectural half of Chapter 4. What Ghost lacks is the dissertation’s detector: there is no LLC estimation, no stagewise-transition monitor, nothing that fires when the policy’s effective truth judgments change. Ghost’s physics-informed priors *constrain* learning but do not *instrument* it.

The honest timescale claim. The dissertation is explicit that this layer is a *slow* assurance mechanism — it quarantines a capability between drives; it cannot prevent a hallucinated command in the microsecond loop. Ghost’s audit infrastructure inherits exactly this limitation. Conflating it with the real-time veto would overstate the result.

Standing: governance/attestation scaffolding *Proposed*; the LLC/SGLD detector *missing and Established-elsewhere* — a known technique Ghost should adopt rather than invent.

5.3.6 Cross-cutting B — The distributed extension (out of scope for the stated goal)

Ghost’s Byzantine fault tolerance ($\lfloor (n-1)/3 \rfloor$ tolerance, 175-step protocol, $1.5\mu\text{s}$ consensus) and population-balance scalability ($O(1)$ via moment closure) are the portfolio’s most distinctive competitive cells. They are also **largely orthogonal to the dissertation’s goal**, which is explicitly the on-road *single-vehicle* case. The dissertation in fact *narrows* the swarm-derived analog-veto principle down to a single vehicle; Ghost’s distributed work is the natural path *back out* to the multi-agent setting (V2X, fleet coordination).

The consolidation’s recommendation is therefore to **decouple** these. The single-vehicle architecture of refusal stands on Stages 0–3 plus Cross-cutting A. Byzantine consensus and population governance are a *second program* — the fleet-scale generalization — that should not be load-bearing for the single-vehicle certification argument, both because they are unproven on driving data and because importing a distributed-systems dependency into the safety-critical guard path would violate the dissertation’s freedom-from-interference posture.

Standing: Byzantine FT and $O(1)$ scalability *Proposed/Notional* and *out of scope* for the immediate goal; valuable as a sequenced follow-on.

5.4 The consolidated architecture

```

+=====+
| ARCHITECTURE OF REFUSAL -- Ghost assets re-indexed by constraint flow |
| (admissible-command set contracts monotonically downward)           |
+=====+
|
| STAGE 0 EPISTEMIC GATE          Ghost: physics-informed margin      |
| trusted scalar phi, one-sided   Owed: one-sidedness proof, safe-side |
| | tightening-only                rounding (Stage 1)                  |
| v                               |
| STAGE 1 NUMERICAL SUBSTRATE      Ghost: GAP -- specify Posit/quire or |
| format = safety variable         interval arithmetic on EPU          |
| | no optimistic rounding        |
| v                               |
| STAGE 2 CONTROL NARROWING        Ghost: MPC-dominant + compositional  |
| CBF/RTF layering; risk-monotone  hyperbolic+ODE reachability         |
| | actuation algebra (clamp,     Owed: rate-clamp-last + exactness    |
| v   rate-last, intersection)    theorem for non-box geometry         |
|
| STAGE 3 REFLEXIVE VETO          Ghost: EPU -- combinational only,    |
| hardwired, fail-closed,         ASIL-D by construction, FPGA        |
| | microsecond timing contract  witness  <= strongest match         |
| v                               |
| ===== ACTUATOR INTERFACE -- physics enforces =====           |
|
| PARALLEL POLICY GOVERNANCE      Ghost: attestation/audit scaffolding  |
| (slow loop, between-drive)      Owed: LLC/SGLD drift detector         |
|
| DEFERRED DISTRIBUTED EXTENSION  Ghost: BFT + population 0(1)         |
| (fleet / V2X, separate program) Scope: NOT in single-vehicle path    |
+=====+

```

The reframe is visible in one fact: of Ghost’s seven “unique cells,” the two that anchor the consolidation (EPU compute, and the MPC/compositional-safety pairing) were rated only “Very High” and “developing” in the competitive matrix, while the two rated “impregnable” (Byzantine FT, sub-5ns latency-as-headline) are respectively *deferred* and *reabsorbed into the veto’s timing contract*. Competitive value and structural value are not the same ranking.

5.5 Provenance ledger

5.6 The verification program (action-oriented roadmap)

Sequenced by the dissertation’s open obligations, not by Ghost’s competitive phase gates. Each item closes a specific seam identified above.

Item 5 deserves emphasis: it is the dissertation’s named central empirical question (a trusted scalar that always reports maximum risk satisfies every safety clause and parks the car forever). Ghost’s portfolio is silent on liveness. No amount of veto rigor substitutes for proving the envelope still permits driving. This is where the consolidated program’s risk concentrates.

5.7 Where the consolidation strains

- **The convergence at Stage 3 is real; the breadth claim is not.** Ghost’s EPU genuinely satisfies the veto’s requirements, and the independent derivation from both certification and silicon directions is strong evidence. But the positioning documents claim the EPU as a general decision substrate. The dissertation’s discipline forbids that: the exhaustively-verifiable

Table 5.1: Ghost portfolio provenance

Asset (positioning-doc claim)	Corpus standing	Role in consolidated solution
Physics-informed priors (patents 11,983,889 / 11,971,958)	Established	Raw material for trusted scalar φ
Compositional hyperbolic-reachability + ODE safety	Proposed (design)	φ witness + Stage-2 barrier
MPC-dominant, equation-appropriate control	Established	Stage-2 narrowing engine
Risk-monotone clamp ordering (rate-last)	Not in corpus	Stage-2 actuation algebra
EPU: combinational-only, ASIL-D-by-construction	Proposed; FPGA witness	Stage-3 analog veto
$< 5\text{ ns}$ / $< 200\text{ pJ}$ EPU figures	Notional (FPGA, not ASIC)	Satisfies timing-contract inequality
Numerical-format conservatism	Gap	Stage-1 substrate
Attestation / audit / continuity tags	Proposed	Governance scaffolding (slow loop)
LLC / stagewise drift detector	Absent (Established elsewhere)	Governance trigger
Byzantine FT (175-step, $1.5\mu\text{s}$)	Proposed/Notional	Deferred fleet extension
Population $O(1)$ moment closure	Proposed (theoretical)	Deferred fleet extension

Table 5.2: Verification roadmap

#	Obligation	Closes	Effort class
1	Specify safe-side numeric format on the EPU checkers	Stage-1 gap	Specification
2	Prove φ is a valid one-sided lower bound for the chosen safe set	Stage-0 obligation	Theory
3	Prove clamp-ordering exactness for the <i>non-box</i> actuator geometry (friction circle)	Stage-2 obligation	Theory
4	Demonstrate fail-closed EPU veto on FPGA under the timing contract	Stage-3 grounding	Engineering
5	Establish the liveness frontier — prove the gate is not so conservative it forbids ordinary driving	Central open question	Empirical
6	Add an LLC/SGLD drift detector behind the attestation logs	Cross-cutting A gap	Engineering + theory
7	Re-scope BFT/population work as an explicit, <i>decoupled</i> fleet program	Scope hygiene	Governance

layer must stay small and stateless. The consolidation buys the EPU as the *enforcer* and explicitly declines it as the *planner*.

- **Two of Ghost’s four “impregnable” cells leave the building.** Byzantine FT and population scalability are the portfolio’s proudest competitive claims and are nearly irrelevant to the single-vehicle goal. Honesty requires saying so; forcing them into the safety path would import a distributed dependency exactly where freedom-from-interference is most needed.
- **The headline numbers are notional and must travel as such.** $< 5\text{ ns}$, $O(1)$, $1.5\text{ }\mu\text{s}$, $296,000\times$ — every one is FPGA-stage or theoretical by the corpus’s own status columns. They are useful as *requirement-satisfaction witnesses*, never as measured device properties. This is the dissertation’s “PICAPD must not masquerade as built” warning applied to Ghost’s own ASIC.
- **The epistemic front-end is the real frontier.** Ghost is rich at the actuator and poor at the percept. The trusted scalar’s one-sidedness, the numerical substrate, and the liveness frontier — the three things that make the chain *both* safe and useful — are precisely the things the four positioning documents do not address. The consolidated solution is therefore not “Ghost already has it.” It is “Ghost supplies the enforcement spine; the perception-side guarantees are the dissertation’s contribution to make, and the open program.”

5.8 Conclusion

Consolidated, Ghost’s portfolio is best understood not as a competitor in a 576-cell tensor space but as the **enforcement substrate for an architecture of refusal**. Its combinational, ASIL-D-by-construction EPU is an independent rediscovery of the dissertation’s analog-veto requirements and is the solution’s strongest, most defensible link. Its MPC-dominant compositional control supplies the narrowing layer above. Its attestation infrastructure supplies the slow-loop governance scaffold. What it does not supply — the one-sided trusted scalar, the safe-side numerical format, the online drift detector, and above all a proof of *liveness* — is exactly the dissertation’s owed contribution, and exactly the program worth funding.

The wager is unchanged from the goal document, only now resourced: make autonomous driving certifiable rather than tested, by proving not that the stack never errs but that when it does, the error has nowhere unsafe to go — with Ghost’s silicon as the layer that makes “nowhere” electrical rather than merely intended.

Consolidated Solution — A Provably Bounded Autonomy Stack via Constraint Propagation: From Project Findings to an Executable Program of Work

6.1 Scope, inputs, and the discipline this document inherits

This solution consolidates two project artifacts that operate at different altitudes:

1. **The synthesis** (*Constraint Propagation from Epistemology to Actuation*), which states the goal and supplies a four-link safety chain — epistemic, numerical, control-theoretic, electrical — together with an explicit ledger separating *established* foundations from *proposed* contributions and *unverifiable* constructs.
2. **The coordination artifact** (`GroupCoordinationFramework.java`), a trust- and memory-aware multi-agent layer that performs group formation, task allocation, weighted consensus, and slow-loop trust adjustment.

The central act of consolidation is to recognize that the second is an instance of the first’s *digital coordination layer*. The framework negotiates *what to do*; by itself it provides no structural safety bound, because every guarantee it offers (its `QUORUM`, `TRUST_THRESHOLD`, and `TIMEOUT` rules; its `WeightedConsensusProtocol`; its negative-trust failure handling) is software-enforceable and therefore defeasible by exactly the hallucination the goal is concerned with. The solution subordinates the framework beneath the constraint chain: its outputs become *proposals* that must survive a trusted-scalar gate, a risk-monotone actuation algebra, and an analog veto before reaching an actuator.

This document carries forward the source’s most important methodological commitment: **the seam between what is known and what is claimed is never blurred**. Established results are cited as foundation. Proposed contributions are flagged as hypotheses with explicit verification obligations. Unverifiable constructs are demoted to design metaphors or notional requirements sketches and never presented as built.

6.2 The goal, restated as an engineering specification

The goal is not “make perception more accurate.” It is a **monotonicity property over admissible actuator commands**:

As control flows from perception toward actuation, and as estimated risk rises, the set of admissible actuator commands may only contract, never expand — and at the final pre-actuation stage this contraction is enforced by physics, not software.

Restated as acceptance criteria a build can be tested against, the solution must guarantee that:

- **G1 (one-sided error).** Every safety-bearing scalar errs only in the conservative direction; it may underclaim safety, never overclaim it.
- **G2 (antitone admissibility).** The admissible-command set is an antitone function of estimated risk — higher risk, strictly smaller set.
- **G3 (non-expansive enforcement).** Projection of any proposed command onto the admissible set is firmly non-expansive (1-Lipschitz), so the enforcer cannot amplify an oscillation injected upstream.
- **G4 (physical fail-closed).** On corrupted data, a missed deadline, or a tampered signal path, an independent hardwired layer forces the minimal-risk envelope within a bounded latency, beneath any software fault.
- **G5 (legible policy).** Any online change in what a learned policy “knows” is detectable and recordable as a challengeable event, not silently absorbed.
- **G6 (conjunctive release).** Actuation is authorized only if every discharge obligation is simultaneously met; the *absence* of a discharged guarantee is itself a fault.

G1–G4 are real-time and structural. G5 is a slower assurance layer. G6 is the integrating gate. The remainder of this document specifies how each is built and verified.

6.3 Solution architecture — the constraint-propagation chain

The architecture is a single pipeline in which each stage can only narrow what the next is permitted to do. Each stage is presented with its established foundation, the mechanism added, and the guarantee it discharges.

The critical reading of S2 is the consolidation’s pivot and is developed next.

6.4 Where the coordination framework fits — and where it must be subordinated

`GroupCoordinationFramework` is a faithful, useful implementation of S2’s *digital negotiation* role. It maps onto the architecture as follows:

- `TrustBasedGroupFormation`, `OptimalTaskAllocation`, and `WeightedConsensusProtocol` together negotiate *what to do* — the “intelligence proposes” half of the analog-veto principle.
- `handleCoordinationFailure`, which applies negative trust deltas across the group on failure, is a **slow-loop assurance** mechanism. It operates on the same timescale as the learning-governance detector of S5, not on the microsecond timescale of S4. The consolidation places it in the S5 band deliberately; treating it as a real-time safety control would overstate it.
- The coordination rules `QUORUM`, `TRUST_THRESHOLD`, and `TIMEOUT` are precisely the kind of *software* gate the goal warns against relying on. They are necessary for sane coordination but provide no structural bound, because a compromised or hallucinating stack can satisfy or bypass them in software.

Three modifications subordinate the framework correctly rather than discarding it:

1. **Demote consensus to a proposal.** The output of `coordinateExecution` / `achieveConsensus` must be typed as a *proposed command*, never an actuator command. It enters S3 as the argument to a projection, not as an authority to move.

Table 6.1: Constraint-propagation chain stages

Stage	Function	Established foundation	Added mechanism
S0 — Sensing & numerics	Produce honest measurements	MEMS noise floors / vibration rectification; DVS, single-photon LiDAR; Posit-with-quire vs IEEE-754 EKF divergence; symplectic (Verlet) prediction	Mandate a numerical format whose rounding error is bounded on the <i>safe</i> side for every safety scalar
S1 — Fidelity gate	Decide whether perception output may authorize action	Barrier-margin / causal-sufficiency witnesses; singular-MDL floor on lossy compression	Trusted scalar φ (Contribution I)
S2 — Coordination / planning	Negotiate <i>what to do</i>	Layered RoM→FoM control; recurrent tracking functions; robust/adaptive & neural/gameplay CBFs; Poisson safety fields	Subordinate the existing GroupCoordination-Framework as a proposer whose consensus and allocations are advisory, not authoritative
S3 — Actuation algebra	Convert risk into a command envelope and project the proposal into it	Convex projection; RSS/SFF envelopes; ODD-transition behavior	Risk-monotone action algebra , intersection-as-meet, asymmetric hysteresis, rate-clamp-last (Contribution II)
S4 — Analog veto	Enforce the envelope beneath software	Guarded-Swarms analog-veto principle; lockstep RISC-V ASIL-D; FPGA jitter-bounded kernels	Timing-contracted hardwired guard , fail-closed to minimal-risk envelope (Contribution III)
S5 — Learning governance	Keep an updating policy legible	LLC + stagewise phase transitions (SGLD-estimable); implicit-bias / “price of implicit bias”	Anti-silent-drift proposer/verifier/adjudicator pattern (Contribution IV bridge)
S6 — Conformance gate	Authorize release of a command	Standards conjunction (§6)	Fail-closed conformance checklist (Contribution IV)

2. **Insert the trusted scalar at the framework boundary.** Each agent contribution aggregated in `aggregateResults` must carry a qualified φ (S1). The aggregation algebra must be restricted to the **conservative join** — the framework’s current implicit averaging of contributions is an anti-pattern under G1 and must be replaced with a meet/most-conservative rule on any safety-bearing field.
3. **Make the framework fail *toward* S4, not toward last-good.** `CoordinationException` paths currently remove the task and update trust; in the consolidated stack they must additionally raise the S4 minimal-risk envelope. The framework’s failure handling becomes a *trigger* for the analog veto, not a substitute for it.

The net effect: the framework retains its real value — trust-weighted multi-agent negotiation,

memory, administrative history — while its authority to actuate is removed and relocated to a layer that physics, not policy, enforces.

6.5 Component specifications (the proposed contributions as build items)

These are the dissertation’s original proposals, restated as build specifications with their verification obligations attached. None is established prior art; each is a hypothesis the program must defend.

6.5.1 Trusted scalar φ (S1) — *proposed*

A single qualified scalar that answers “can the discarded detail only ever shrink, never enlarge, the admissible action set?” Five-part contract: (i) denotes a genuine safety quantity (barrier margin or causal-sufficiency witness, not a PSNR proxy); (ii) bounded and saturating; (iii) **overestimates** risk on error; (iv) feeds only constraint-*tightening* maps; (v) continuously re-qualified, escalating monotonically to maximum risk on any conformance failure.

Forbidden: treating L2 accuracy as the safety criterion; letting “high fidelity” authorize a more aggressive action; **averaging** a trusted scalar with an untrusted one.

Verification owed: (a) a proof that the chosen φ is a valid one-sided bound for the given safe set under stated assumptions; (b) a numerical-format argument (S0) that φ cannot round in the optimistic direction. Until both hold, the contract is aspirational.

6.5.2 Risk-monotone actuation algebra (S3) — *proposed*

A family of admissible-command sets indexed by a normalized risk band, antitone in risk. Each set is a closed, convex, axis-aligned box, so projection is unique and firmly non-expansive. Implement as a two-stage clamp — magnitude, then **rate last** — so a risk spike yields bounded jerk (graceful distance-decay) rather than an actuator step-lurch. Multiple envelopes compose by **intersection (meet)**, never union or average. Band switching uses **asymmetric hysteresis**: tighten instantly on the up-threshold, loosen only below a strictly lower down-threshold after a dwell.

The contribution is the ordering. The defensible, non-obvious claim is that clamp ordering is a safety property: rate-last ratchets into the tighter envelope at maximal bounded jerk; magnitude-last would force an unbounded step.

Verification owed: the **exactness theorem** — that sequential magnitude-then-rate clamping equals the true Euclidean projection — holds for axis-aligned boxes only. Real coupled steering/acceleration limits and friction circles are not axis-aligned. The program must either extend the theorem to the relevant non-box geometry or restrict claims to the regime where the box approximation is itself conservative.

6.5.3 Timing-contracted analog veto (S4) — *proposed, on real foundation*

An independent analog layer enforcing S3 under the timing contract: intervention latency $\ll$ sensor-consistency window $\ll$ digital decision window. Hardwired comparators and latches, no programmable logic in the guard path, so a compromised stack cannot reach the actuator first. On corrupted data, missed deadline, or tampered signal path it forces the minimal-risk envelope (maximal safe braking, zero positive acceleration) — **fail-closed**, not last-good coasting.

Notional hardware. The “PICAPD ISA” is treated strictly as a *requirements sketch* — privileged non-user-writable safety registers, deterministic bounded latency, architectural rollback to a known-safe configuration on a violated safety bit. It is **never** cited as an existing device. The realizable near-term substrate is FPGA-accelerated jitter-bounded kernels over a lockstep RISC-V (ASIL-D) path.

6.5.4 Conformance checklist (S6) — *proposed integrating artifact*

A single fail-closed release gate. A reconstruction may drive actuation only if, simultaneously: (1) φ is a qualified trusted scalar; (2) every fused input passed a *pre-fusion* admissibility test and carries provenance; (3) RoM→FoM tracking error is within its recurrent bound; (4) the command lies in (or is non-expansively projected into) the intersection of all active envelopes, rate-clamp last; (5) no averaging, union, or less-conservative-meet appears anywhere on a safety-bearing path; (6) the retrieval that produced the reconstruction respected its compute/latency budget; (7) no online policy update silently revised ground truth; (8) any failed check has already placed the system in the minimal-risk envelope. The clauses are **conjunctive**.

6.6 Verification and certification argument

The payoff is a certification posture road-miles alone cannot reach: SOTIF compliance becomes a *structural* claim about what an error can do, not a *statistical* claim about how rarely the stack errs.

Table 6.2: Standards crosswalk for the consolidated stack

Standard	What it demands	How the consolidated stack answers it
ISO 26262 (ASIL)	Mitigate E/E faults; freedom from interference	S4 hardware veto, deterministic bounded latency, lockstep substrate, S0 numerical resilience
ISO 21448 (SOTIF) / UL 4600	Bound the “unknown unsafe” with no hardware fault	S1 trusted-scalar gate + S3 algebra make unsafe commands physically unrepresentable
IEEE 2846 / RSS / SFF	Minimum reasonable assumptions; behavior across ODD transitions	S3 rate-clamp-last bounded-jerk entry into stricter RSS envelopes; asymmetric hysteresis
AEC-Q100 / ISO 16750-3	Thermomechanical / vibration survivability	S0 vibration & noise floor sets the real lower bound on any safety threshold; format/precision choices

The decisive claim the chain licenses: *even under severe algorithmic hallucination in perception or planning, the actuation-boundary forms force the vehicle into a provable Minimum Risk Condition* — and do so without violent braking at ODD transitions, because the same rate-clamp-last discipline that prevents lurching is what carries the vehicle smoothly into the more conservative RSS parameters that fog or rain demand.

6.7 Implementation roadmap

Phased so that each phase produces an independently defensible result and the load-bearing proofs precede the claims that depend on them.

1. **Phase A — Substrate and gate (S0, S1).** Establish the bounded-on-the-safe-side numerical format for all safety scalars (Posit-with-quire vs IEEE-754 baseline, EKF-divergence regression test). Define φ for one concrete safe set and discharge verification obligation 4.1(a)+(b). *Exit:* a φ that is a proven one-sided bound, computed in a non-optimistically-rounding format.

2. **Phase B — Actuation algebra (S3).** Implement the antitone box family, non-expansive projection, rate-clamp-last, intersection-meet, asymmetric hysteresis. Prove the exactness theorem for the axis-aligned case and characterize the non-box regime (4.2). *Exit:* a projection enforcer with a proven exactness/conservatism property and bounded-jerk band transitions.
3. **Phase C — Analog veto (S4).** Build the hardwired guard on FPGA/lockstep-RISC-V; validate the latency ordering of the timing contract on the bench; confirm fail-closed behavior on injected corruption, deadline miss, and signal tamper. *Exit:* measured bounded intervention latency beneath the digital decision window.
4. **Phase D — Subordinate the coordination framework (S2→S3).** Apply the three modifications of §3: type consensus outputs as proposals; insert φ at the aggregation boundary and replace averaging with conservative join; route `CoordinationException` to raise the S4 envelope. *Exit:* the framework can no longer issue an actuator command except through the projection enforcer.
5. **Phase E — Learning governance (S5).** Stand up LLC/SGLD detection of stagewise transitions; implement the proposer/verifier/adjudicator anti-silent-drift log; treat optimizer-bias-vs-threat-geometry alignment as a first-class design constraint. *Exit:* online capability changes are logged and quarantinable between drives.
6. **Phase F — Conformance gate and certification crosswalk (S6, §5).** Wire the conjunctive checklist as the release gate; assemble the standards evidence package. *Exit:* a fail-closed gate and a SOTIF/UL 4600 structural argument.

6.8 Risk register and open problems

These are the honest failure points; the program’s credibility depends on holding them in view.

- **The exactness theorem is owed, not proven** beyond axis-aligned boxes; real actuator constraints are coupled and non-box. (*Phase B gate.*)
- **The trusted scalar can be vacuously safe.** A φ that always reports maximum risk satisfies every clause and parks the vehicle forever. The unaddressed hard part is **liveness** — proving the gate is not so conservative it forbids ordinary driving. This is the program’s central open empirical question and lives on the same trade-off frontier the robust-CBF adaptation literature already grapples with.
- **Detection latency $\neq$ actuation latency.** S5’s LLC detector runs on slow-loop timescales; S4 runs in microseconds. Governance cannot prevent a hallucinated command in real time — it can only quarantine a *capability* between drives. Conflating the two timescales would overstate the result; the analog veto is the real-time guarantee, learning governance is the slower assurance layer.
- **Notional hardware must not masquerade as built.** Specific cycle counts, opcodes, or register files for “PICAPD” must be presented as *required properties*, not measured ones; the realizable substrate is FPGA/lockstep-RISC-V.
- **Unverifiable citations must be cut or downgraded.** The “Learning Mechanics” coalition framing and its associated preprint identifier could not be confirmed; its underlying components (PDLT, SLT/LLC, implicit bias, equivariance, SAEs) are cited individually instead. Every citation on a load-bearing path is to be independently verified before any submission.

Appendix — Provenance ledger (carried forward)

Table 6.3: Provenance ledger

Claim or construct	Standing	Treatment in this solution
CBF / RoM–FoM layered control / RTF / Poisson safety / neural & gameplay filters	Established	Foundation for S2/S3
MEMS, DVS, quantum sensing, FFT accelerators, neuromorphic, RISC-V ASIL-D, Posit, Verlet, AEC-Q100	Established	Substrate S0; lockstep substrate for S4
PDLT; SLT/LLC; implicit bias; equivariance; SAEs/mech-interp	Established	LLC as S5 detector
ISO 26262, SOTIF/21448, UL 4600, RSS, SFF, rule-books, IEEE 2846	Established	Certification crosswalk §5
Guarded Swarms analog-veto principle	Established (verified publication)	Basis of S4 (Contribution III)
GroupCoordination-Framework (trust/consensus/memory)	Project artifact	Subordinated as S2 proposer; failure paths route to S4
“Trusted Scalar” five-part contract	Not in literature	Contribution I (S1, proposal); obligations 4.1(a)+(b) owed
“Risk-Monotone Action Algebra” /clamp-ordering	Not in literature	Contribution II (S3, proposal); exactness theorem owed
“Constitutional Truth Governance” / “Universal Filtration Primitive”	Not in literature	Design metaphors only (S5); not cited as theory
“PICAPD” ISA (24-core die, 512 event regs, msafety CSR)	Could not be verified	Notional requirements sketch for S4; never a built device
“Learning Mechanics” coalition + associated preprint ID	Could not be verified	Downgraded; underlying components cited individually

Consolidating the EPU Program into a Provably Bounded Autonomous-Driving Solution: Mapping the Queen Bee / Event Processing Unit Findings onto the Constraint-Propagation Thesis

Document type, status, and companions

Document type: Project consolidation / solution architecture.

Status: Synthesis of established project artifacts against the stated dissertation goal.

Companion to: *Constraint Propagation from Epistemology to Actuation* (the goal document); the EPU project corpus (Binary-State-Machine Architecture, Queen Bee Memoryless Model, Autonomous Driving Fortified, Hardware/Validation Roadmaps, Physics-Informed Certification).

7.1 The consolidating claim

The dissertation goal and the EPU project are not two programs of work; they are one architecture described from its two ends. The dissertation specifies, **top-down**, a chain in which the admissible-command set may only contract as one moves from perception toward the wheels, terminating in an enforcement layer that uses physics rather than software. The EPU corpus builds, **bottom-up**, a memoryless binary substrate — the Queen Bee Worker→Manager→Queen hierarchy — whose entire reason for existing is to compress a high-dimensional sensory context into a single committed actuator decision through hardwired, combinational gates.

Consolidated thesis. The Queen Bee EPU *is* the dissertation’s monotone admissible-set contraction made physical. Each tier of hierarchical compression is one link in the constraint chain; the final hardwired decision gate, protected by selective triple-modular redundancy and defaulting to a safe state on any fault, is the dissertation’s “architecture of refusal” instantiated in silicon. The project therefore does not need a new architecture to meet the goal — it needs a disciplined **reframing** of what its existing decision bit *means*, three **targeted extensions**, and an honest accounting of which of its performance figures are *built* versus *projected*.

The remainder of this document discharges that consolidation: it states what the project already delivers against the goal, what must be reframed or extended, what must be proven, and how the existing 18–21-month roadmap absorbs the additional obligations without restructuring.

A note on right-of-way, per the project’s own instruction: where the dissertation’s framing and the project files diverge, the **project files govern the engineering**, and the dissertation

governs the **standard of evidence**. The two are complementary — the project supplies the concrete substrate and numbers; the dissertation supplies the seam between *what is built* and *what is claimed* that the project documentation, in places, blurs.

7.2 The architecture in one frame

DISSERTATION (top-down: set contracts)	PROJECT (bottom-up: built substrate)	
Perception / fused sensing	+-----+-----+	Sensor preproc (GPU/FPGA, ~10 ms)
- fidelity is a *trusted scalar* ϕ	EPISTEMIC LINK	10,000 sensor bits @ 100 Hz
- ϕ may err only toward more risk	+-----+-----+	
	context flows in (no memory fetch)	
Reduced-order safety reasoning	+-----v-----+	WORKER tier (100 units, ~1 ns)
- admissible set begins to shrink	CONTRACTION 1	100 bits -> 100 "object here?" bits
	+-----+-----+	
Risk-banded command envelope	+-----v-----+	MANAGER tier (10 units, ~1 ns)
- antitone: rising risk -> smaller	CONTRACTION 2	100 bits -> 10 zone decisions
- intersection (meet), never union	+-----+-----+	
Pre-actuation projection	+-----v-----+	QUEEN tier + sel. TMR (~1 ns +0.2 ns)
- non-expansive clamp, rate-last	CONTRACTION 3	10 bits -> 1 committed, majority-voted
- fail-closed to minimum-risk set	+-----+-----+	
	binary output flows out (~0.1 ns wire)	
Analog veto (physics, not code)	+-----v-----+	ACTUATOR INTERFACE
- hardwired, sub-us, no programmable logic in the guard path	ENFORCEMENT	Brake/steer cmd; physical lag dominates (brake 50-80 ms, steer 30-50 ms)
	+-----+-----+	

The left column is the goal; the right column is the corpus. The consolidation is the assertion that each box on the left is discharged — fully, partially, or not yet — by the artifact on the right.

7.3 What the project already delivers against the goal

These are links where the built or specified EPU artifact substantively satisfies the dissertation's requirement. They are the program's strongest assets and should be presented as such.

7.3.1 The constraint chain exists physically (Contractions 1–3)

The dissertation's central formal claim is monotonicity: *as one moves toward actuation, the admissible-command set may only contract*. The Queen Bee hierarchy enforces exactly this by construction. Each tier performs irreversible binary compression — 100 bits to 1, then 10 to 1, then 10 to 1 — and a downstream tier can never re-expand information a prior tier discarded, because the discarded bits are not stored anywhere (the architecture is *memoryless*). The 10,000:1 hierarchical compression ratio the project repeatedly cites is, read through the dissertation, a *measure of how far the admissible set has been narrowed* by the time a command is committed. The project's information-physics certification correctly establishes that this ratio is necessarily *lossy* (it exceeds the Shannon lossless bound by orders of magnitude), which is the dissertation's point restated in information-theoretic terms: the chain deliberately destroys degrees of freedom.

7.3.2 The enforcement layer is hardwired and combinational

The dissertation's most load-bearing requirement is that the final narrowing be enforced "by physics rather than software" — a guard path of "hardwired comparators and latches with no programmable logic." The EPU gate-enforcement hardware (the GEU) and the Worker/Manager/Queen gates are specified as exactly this: pure combinational boolean logic evaluating in roughly one nanosecond, not sequential computation over fetched state. The

project’s framing — “*Intelligence proposes; hardware disposes*” in the dissertation, “*binary decision gates replace memory-based state machines*” in the corpus — is the same principle. **This is the single greatest point of alignment**, and it is what licenses the dissertation’s strong certification claim (§5).

7.3.3 Fail-closed default and selective TMR

The dissertation requires that on corrupted data, a missed deadline, or a tampered signal path, the enforcer force the minimal-risk envelope — a *fail-closed* default rather than coasting on the last good command. The EPU autonomous-driving design specifies a safe state (hazard lights + graduated deceleration) and a fault-detection budget under 100 ms, with **selective TMR applied only to the final Queen decision** and majority voting tolerating single-point faults. This is a precise realization of the dissertation’s fail-closed requirement, and the project’s selective-redundancy strategy (TMR on the committed decision only, not the whole chip) is an efficiency refinement the dissertation does not itself supply.

7.3.4 The timing contract is satisfied with enormous margin

The dissertation states the enforcement timing contract as: *intervention latency* $\ll$ *sensor-consistency window* $\ll$ *digital decision window*. The project’s headline finding — a sub-nanosecond-class decision path (≈ 3.4 ns projected on a 28 nm ASIC; ≈ 32 ns measured on FPGA) against actuator lags of 30–80 ms — satisfies this by six to seven orders of magnitude. The project’s own observation that “EPU latency is irrelevant to overall system latency; the physical world dominates” is, in the dissertation’s terms, the statement that **the guard always wins the race to the actuator**, which is the property the timing contract demands.

7.3.5 Physics-certification is the trusted-scalar discipline, generalized

The dissertation’s Contribution I requires that any scalar gating actuation be computed so that it can only err on the safe (more-conservative) side, and that its computation cannot round in the optimistic direction. The project’s Phase-0 physics-informed certification — nine constraint domains, ~ 900 tests, $\geq 95\%$ pass threshold, physics violations treated as objective training signals rather than post-hoc corrections — is a broader instance of the same discipline: it forces the design-time reasoning layer to be *wrong only in physically admissible directions*. The corpus’s insistence that latency, energy, and thermal claims must conserve and couple correctly is exactly the dissertation’s “wrong only in the safe direction,” lifted from the runtime scalar to the design-time co-pilot.

7.3.6 Standards posture

Both documents converge on the same certification crosswalk. The consolidated mapping:

7.4 What must be reframed or extended to fully achieve the goal

The project meets the *spirit* of the goal at every link, but three gaps separate the current corpus from the dissertation’s *provable* claim. None requires a new architecture; each is a targeted modification of the existing one.

7.4.1 Reframe the Queen decision from a binary to a risk-monotone projection (the central reframing)

Current state. The Queen tier emits a single bit: brake / no-brake, change-lane / maintain. The dissertation requires a *risk-banded admissible-command set* with the antitone property —

Table 7.1: Standards crosswalk for the EPU

Standard	Demand	Discharged by (project artifact)
ISO 26262 (ASIL-D)	Mitigate E/E faults; freedom from interference	Selective TMR on Queen tier; < 1 FIT projected; watchdog + confidence bit; deterministic combinational path
ISO 21448 / SOTIF; UL 4600	Bound the “unknown unsafe” with no hardware fault	Memoryless contraction chain makes an unsafe command unrepresentable downstream of the gate it violates
IEEE 2846 / RSS / SFF	Reasonable assumptions; behaviour across ODD transitions	(Owed — see §5.1: requires the banded envelope, not the current binary decision)
AEC-Q100 / ISO 16750-3	Thermomechanical / vibration survivability	Environmental physics certification domain; burn-in test flow; automotive qualification in roadmap Phase 6

rising estimated risk shrinks a continuous envelope, and the committed command is the *non-expansive projection* of the proposed command onto that envelope, with the rate clamp applied last so a sudden risk spike produces bounded jerk rather than an actuator step-lurch.

The gap. A pure binary decision cannot express “smoothly bleed steering and acceleration down over $\sim 1.25\text{--}1.47$ s into a tighter envelope.” It can only switch. This is precisely the behaviour the dissertation identifies as the difference between a *tested* and a *certifiable* ODD transition (the smooth entry into stricter RSS parameters demanded by fog or rain).

The resolution. The hierarchy is retained; the Queen’s *output alphabet* is widened from one bit to a small banded envelope index plus a hardwired clamp stage. The Manager tier already computes zone-level decisions that can be read as risk-band inputs; the Queen’s combinational logic is extended to (a) select the active band by **intersection (meet)** of all active envelope constraints — never union or average, which the dissertation shows destroys the worst-case bound — and (b) drive a two-stage clamp (magnitude, then **rate last**) implemented as additional combinational logic before the actuator pad. This keeps the nanosecond, memoryless, gate-only character intact while making the output a contraction operator rather than a switch. **The project’s worked confusion-matrix validation (TPR/FPR targets) must then be supplemented with an envelope-conformance test**, because correctness is no longer “did it pick the right bit” but “did the committed command lie inside the intersection of active envelopes.”

7.4.2 Separate the analog guard channel from the digital Queen

Current state. The corpus’s Queen tier, though combinational and fast, is still *digital logic on the primary decision die*. The corpus’s “PICAPD ISA” (24-core die, 512-bit event register file, privileged `msafety` CSR) is invoked as if it were a built device with specific cycle counts.

The gap. The dissertation is explicit and correct: the guard path that performs the final veto must contain *no programmable logic*, so that a compromised or hallucinating digital stack — including a faulted Queen — cannot reach the actuator before the guard acts. A digital Queen,

however fast, is part of the stack the guard must be independent of. And the PICAPD ISA, per the dissertation’s verification finding, **could not be confirmed as a built artifact** and must be treated as a *notional requirements sketch*, never cited as measured silicon.

The resolution. Introduce an independent analog/hardwired-comparator veto layer *downstream of* the digital Queen, sitting directly at the actuator interface. Its only behaviours are: pass the Queen’s clamped command through when all guard conditions hold, or force the minimal-risk envelope (maximal safe braking, zero positive acceleration) otherwise. This layer is the dissertation’s adaptation of the verified *Guarded Swarms* analog-veto principle and is the program’s true real-time guarantee. The project’s existing watchdog timer and confidence bit are the natural seed for it, but they must be relocated *out of* the programmable path and hardened into the guard. The FPGA-accelerated, jitter-bounded reflexive layer the project already prototypes (32 ns measured) is the credible near-term substrate; lockstep RISC-V is the credible ASIL-D digital companion. **The PICAPD ISA should be presented throughout as the set of *required properties* — privilege isolation, bounded rollback latency, jitter-bounded execution — that any conforming enforcer must exhibit, not as a chip that exists.**

7.4.3 Add the learning-theory governance overlay for field updates

Current state. The corpus’s continuous-improvement loop updates decision thresholds from fleet data via OTA “configurable registers” or hardware revision, with accuracy projected to climb from 95% toward 99.9%.

The gap. This is exactly the *silent-drift* hazard the dissertation flags. An OTA threshold change “absorbed into the canon” can quietly revise what the policy treats as ground truth, with no challengeable record. The dissertation’s learning-theory bridge supplies the missing instrumentation: a developmental phase transition in a learning component is observable as a coupled regret-drop / Local-Learning-Coefficient jump, and any update that alters effective truth judgments should be logged as a formal, challengeable event and, where warranted, quarantined between drives.

The resolution. Wrap the OTA path in a proposer/verifier/adjudicator separation: the update mechanism *proposes* new thresholds; an independent verifier checks them against the physics-certified constraint database (§3.5) and the envelope-conformance suite (§5.1); an adjudicator either canonizes or quarantines, preserving ambiguity when evidence is insufficient. Crucially, the dissertation’s honest caveat must be carried forward verbatim: **this governance operates on slow-loop timescales and cannot prevent a hallucinated command in real time** — only the analog veto of §5.2 does that. Conflating the two timescales would overstate the result.

7.5 What must be proven (open obligations carried into the program)

Consolidation does not dissolve the dissertation’s open obligations; it assigns them owners and schedules. Four remain genuinely open and must be stated plainly rather than asserted away.

1. **The exactness theorem beyond box constraints.** The clamp-ordering guarantee (rate-last = bounded jerk, sequential clamp = true Euclidean projection) holds for axis-aligned convex boxes. Real actuator limits — coupled steering/acceleration, the friction circle — are not axis-aligned. The program owes either a proof that the result extends to the relevant non-box geometry, or an explicit restriction of the claim to the regime where the box

approximation is itself conservative. **This is the binding mathematical obligation on §5.1.**

2. **The liveness / feasibility frontier.** A guard that always reports maximum risk satisfies every safety clause and parks the vehicle forever. The hard, unaddressed problem is proving the gate is *not so conservative that it forbids ordinary driving*. This is the program’s central empirical question and is the same trade-off the robust-CBF adaptation literature already grapples with. The envelope-conformance suite (§5.1) must therefore be paired with a liveness suite measuring false-veto rate against a drivability baseline.
3. **The numerical one-sided-bound argument.** Any scalar gating actuation must be proven (a) a valid one-sided bound under stated assumptions, and (b) computed in a format whose rounding error cannot land on the optimistic side. The project’s own finding that *numerical format is a safety variable* (IEEE-754 single precision can drive an extended Kalman filter to diverge where Posit-with-quire tracks the double-precision reference) is the tool: any safety-bearing scalar in the reframed Queen must use a conservatively-rounding format.
4. **Notional-versus-built honesty across the entire corpus.** The project’s specific figures — 3.4 ns decision latency, 116 pJ/decision, < 1 FIT, \$2.53/die, 79.1% yield, 95→99.9% accuracy — are **design targets and projections, not measurements of fabricated silicon**. The only measured number in the corpus is the FPGA demonstration (~32 ns). The dissertation’s discipline is non-negotiable here: presenting projected cycle counts or FIT rates as measured properties would be a certification failure. Every such figure in downstream documentation must be labelled *projected* until validated against the Phase-6 fabrication and qualification results.

7.6 The consolidated solution architecture

7.6.1 The integrated stack

```

Fused sensing --> Physics-certified perception/reasoning (design-time trusted-scalar, S3.5)
                  | emits a conservatively-rounded risk scalar phi (S5.3)
                  v
Worker tier     --> Contraction 1: memoryless 100->1 binary compression (S3.1, built)
                  v
Manager tier    --> Contraction 2: zone decisions = risk-band inputs (S3.1, built; reused S5.1)
                  v
Queen tier     --> Contraction 3: band selection by MEET, mag-then-RATE-LAST clamp (S5.1, reframe)
                  | selective TMR on the committed command (S3.3, built)
                  v
Analog veto    --> Independent hardwired guard, no programmable logic, fail-closed to MRC (S5.2, extend)
                  v
Actuator       physical lag dominates; guard has already won the race (S3.4, built)

-- slow loop, off the real-time path --
Field updates  --> Proposer / verifier / adjudicator; LLC-jump detector; quarantine (S5.3, extend)
    
```

7.6.2 The release gate (consolidated conformance checklist)

A reconstruction may drive actuation only if **all** of the following hold simultaneously; the *absence* of any discharged guarantee is itself a fault that forces the minimal-risk envelope:

1. the gating scalar φ is a qualified, conservatively-rounded one-sided bound (§3.5, §5.3);
2. every fused input passed a pre-fusion admissibility test and carries provenance;
3. the reduced-order $\rightarrow$ full-order tracking error is within its recurrent bound;

4. the committed command lies in (or is non-expansively projected into) the **intersection** of all active envelopes, **rate-clamp last** (§5.1);
5. no averaging, union, or less-conservative meet appears anywhere on a safety-bearing path;
6. the analog guard channel is live, independent, and within its timing contract (§5.2);
7. no online policy update silently revised ground truth (§5.3);
8. any failed check has already placed the system in the minimal-risk condition (§3.3).

The decisive consolidated claim this gate licenses is the dissertation’s, now grounded in the project’s substrate: **even under severe algorithmic hallucination in perception or planning, the actuation-boundary forms force the vehicle into a provable Minimum Risk Condition** — and the memoryless hierarchy guarantees there is no stored optimistic state for a hallucination to exploit downstream of the gate it violates. SOTIF compliance thereby shifts from a statistical claim about how rarely the stack errs to a structural claim about what an error can *do*.

7.7 Roadmap consolidation

The dissertation’s additional obligations fold into the project’s existing 18–21-month roadmap without restructuring. The mapping:

The mathematical obligation (§6.1, the exactness theorem beyond box constraints) and the empirical obligation (§6.2, the liveness frontier) are the two items that are *research*, not *engineering*, and should be run as parallel workstreams from Month 0 rather than scheduled into a build phase — they gate the validity of the claim, not the fabrication of the chip.

7.8 Provenance ledger (carried forward, unblurred)

The consolidation preserves the dissertation’s discipline of never blurring the seam between what is known, what is built, what is claimed, and what is unverified.

7.9 Conclusion

The consolidation reaches a clean result: **the EPU program already builds the architecture the dissertation goal demands, but currently describes its output too narrowly and its evidence too confidently**. The Queen Bee hierarchy is the monotone constraint chain; the combinational gates and selective TMR are the fail-closed enforcer; the timing budget wins the race to the actuator with orders of magnitude to spare; the physics-certification is the trusted-scalar discipline in another guise.

What stands between the corpus and the dissertation’s *provable* claim is not a redesign but three disciplined moves and four honest acknowledgements: widen the Queen’s decision from a bit to a rate-clamped risk-monotone projection; lift the final veto out of programmable logic into an independent analog guard; govern field updates so capability changes are challengeable rather than silent; and prove the exactness and liveness results while labelling every projected figure as projected until silicon says otherwise.

The wager of the consolidated program is the dissertation’s, now resting on a substrate the project has already specified: autonomous driving becomes *certifiable* rather than merely *tested* — not by proving the stack never errs, but by proving, in hardwired logic that no software fault can reach past, that when it does err the error has nowhere unsafe to go.

Table 7.2: Roadmap consolidation

Existing phase	Existing scope	Consolidated addition
Phase 0 (Months $-3 \rightarrow 0$)	Physics certification, 9 domains, certified LLM co-pilot	Add the one-sided-bound and conservative-rounding constraint domains (§5.3); add the envelope-conformance and liveness test families (§5.1)
Phase 1 (Months 0–3)	Microarchitecture spec	Widen Queen output alphabet to banded envelope + clamp (§5.1); spec the independent analog guard as a distinct block (§5.2)
Phase 2 (Months 4–7)	RTL + verification	Add meet-not-union and rate-clamp-last assertions to formal verification; verify guard independence (no programmable path to actuator)
Phase 3 (Months 8–10)	Physical design	Place the analog veto downstream of and isolated from the digital Queen die; conservatively-rounding arithmetic for φ
Phases 4–5 (Months 11–15)	Fab + validation	Replace projected figures with measured ones ; the FPGA 32 ns result is the only validated latency until ASIC silicon returns
Phase 6 (Months 16–18)	Production + qualification	AEC-Q100 / ISO 16750-3; ISO 26262 ASIL-D dossier built around the conformance gate of §7.2
Continuous	OTA improvement loop	Wrap in proposer/verifier/adjudicator governance; LLC-jump quarantine (§5.3)

Table 7.3: EPU provenance ledger

Element	Standing	Treatment in this consolidation
Queen Bee memoryless hierarchy; Worker/Manager/Queen contraction	Project design, internally specified	Accepted as the physical realization of the constraint chain (§3.1)
Hardwired combinational gates; selective TMR; fail-safe state	Project design	Accepted as the enforcement layer and fail-closed default (§3.2–3.3)
Physics-certification (9 domains, $\geq 95\%$)	Project method	Accepted as the trusted-scalar discipline generalized (§3.5)
Guarded Swarms analog-veto principle	Established (verified publication)	Basis for the independent analog guard (§5.2)
CBF / RoM–FoM / RTF / Poisson / neural & game-play filters; SLT/LLC; standards landscape	Established literature	Accepted as foundations the program builds on
Trusted-scalar contract; risk-monotone algebra; clamp-ordering	Not in literature	The program’s own proposed contributions; exactness theorem owed (§6.1)
3.4 ns / 116 pJ / < 1 FIT / \$2.53 die / 79.1% yield / 95→99.9% accuracy	Projected, not measured	Labelled projected until Phase-6 validation (§6.4)
FPGA ≈ 32 ns latency	Measured (demonstration)	The only validated timing figure to date
“PICAPD” ISA (24-core, 512-bit ERF, msafety CSR)	Could not be verified	Treated as a notional requirements sketch, never as built silicon (§5.2)
“Learning Mechanics” umbrella + associated preprint ID	Could not be verified	Downgraded; underlying components (SLT, LLC, implicit bias) cited individually

Consolidating the Project into a Provably Bounded Autonomy Stack: How the Biomass-Gasification Framework, the PICAPD/Hardware ISA, and the Critical Review Compose into the Constraint-Propagation Goal

8.1 The goal, restated as an engineering specification

The attached synthesis sets a single target: safety in autonomous driving should be engineered as an **unbroken chain of constraint propagation**, in which every degree of freedom available to a fallible perception-and-planning stack is narrowed, stage by stage, until unsafe actuator commands are *physically unavailable* at the actuator interface — not merely discouraged. Formally, the admissible-command set may only contract as one moves from perception toward actuation and as estimated risk rises, with the final contraction enforced by physics rather than software.

The synthesis advances four proposed contributions to reach that target — a trusted-scalar fidelity gate (I), a risk-monotone actuation algebra (II), a timing-contracted analog veto (III), and a conformance checklist that binds them (IV) — and is candid that, *as written*, these are organizing devices in need of grounding. It treats the “PICAPD” architecture as a notional sketch and flags several constructs as unverified.

The central finding of this consolidation reverses that posture. **The grounding the synthesis asks for is already present in the project** — distributed across the gasification dissertation, the LaTeX paper draft, the PICAPD instruction-set architecture, and the Hardware ISA. The remainder of this document shows that the four “proposed” contributions are not aspirational: each has a worked, partially validated instantiation somewhere in the corpus. The task is to recognize that the four documents describe *one machine at four altitudes*, and to wire those altitudes into the constraint chain the goal demands.

8.2 The unifying discovery: a conservation-manifold compiler

The four corpora look unrelated — thermochemistry, a RISC-V derivative, a peer-review rebuttal, a safety thesis. They are not. They share one mechanism, which is precisely the mechanism the safety goal requires but does not yet possess.

The mechanism. Take a high-dimensional, noisy, conservation-constrained state. Reduce it not by averaging but by *projecting onto a manifold that encodes the conservation law structurally*,

then read off a small set of qualified scalars whose relationship to the quantity of interest is **monotone** and whose computation **cannot violate the constraint by construction**.

In the gasification work this is literal and validated. Compositional state lives on the Dalton simplex $\sum y_i = 1$; the framework re-coordinates it as Pythagorean geometry $\sin^2 \theta + \cos^2 \theta = 1$ — the “open-water” transformation — so the conservation constraint becomes the geometry of the space rather than a penalty added to a loss. Dimensional reduction (canonical correlation analysis, Fibonacci sequence encoding, neural “computational laboratories”) feeds a complete elliptic integral $K(m)$ computed by the arithmetic-geometric mean, which is mapped through an **ellipap**-style module to a transfer function

$$H(s) = k \frac{\prod_i (s - z_i)}{\prod_j (s - p_j)}.$$

The transfer function’s pole–zero–gain topology *enforces* mass and energy conservation through passivity ($\text{Re}(p_j) < 0$), stability, and causality (denominator degree $\geq$ numerator degree). Conservation holds **by construction, not by post-hoc correction or soft penalty**. From $H(s)$ the framework extracts the qualified scalars:

- $\xi = \ln(\Lambda_S/\Lambda_G)$, a continuous log-deviation that stays $O(1)$ even though the raw gain k spans 10^{-47} to 10^{-60} ; **monotone with product-gas yield**.
- $S = \mathbf{1}[n_z \neq n_p] \in \{0, 1\}$, a discrete structural parity read directly off pole–zero topology; **monotone with carbon-conversion efficiency**.
- sct , a “specific computational time” used as a non-equilibrium proxy.

This is exactly the object the safety thesis calls a *trusted scalar*: a single bounded, saturating, monotone quantity that summarizes a fallible high-dimensional reduction and is tied structurally to a conserved invariant. The gasification program demonstrates the pattern is realizable and predictive ($R^2 > 0.78$ across all outputs, $R^2 = 0.87$ for CCE, at 10^5 – $10^6 \times$ the speed of CFD).

The same mechanism is already in silicon. The PICAPD ISA — *Physics-Informed Computing for Autonomous Population Dynamics* — implements these exact primitives as instructions: **AGM.ITER**, **ELI.COMP**, and **TXF.PASS** (transfer-function passivity) in the AGM/Elliptic class; **MOM.CALC/MOM.REAL** for moments and realizability; **VERLET/HAMILT/LEVAL** for symplectic variational mechanics. It exposes a dedicated **Safety/Constraint** instruction class (**CONS.CHK**, **RES.BUDG**, **BYZ.CONVS**, **TMR.VOTE**, **SAFE.ROLL**), a privileged, non-user-writable **msafety** control/status register (0x807), a **ConstraintViolation** trap (code 9), and a Sensor-Fusion class (**SFSPU.SYNC/KF/IPS/PROJ**) whose timing is tuned to **automotive sensors** — IMU at 10 kHz, camera at 60 Hz, LiDAR at 10 Hz, GPS at 1 Hz, with an extended Kalman filter update. The Hardware ISA gives the safety primitives bounded latencies: **CONS.CHK** at 2.7 ns, **SAFE.ROLL** at 28.4 ns (state restore from the write-ahead log), and triple-modular-redundancy majority voting that is single-fault tolerant and Byzantine-detecting.

So the picture resolves: the gasification dissertation is the **mathematical proof of concept** for a conservation-manifold compiler that emits monotone trusted scalars; the PICAPD/Hardware ISA is the **silicon that runs that compiler** and already carries the hard-wired safety machinery the analog veto requires; the critical review is the **adversary** that tells us which of these claims will survive defense and how to harden them; and the synthesis is the **target architecture** that says where each piece must sit in the chain from epistemology to actuation.

The consolidation, then, is a single sentence with the rest of the document as its proof:

The validated conservation-manifold compiler from the gasification work is the missing instantiation of the trusted scalar; the PICAPD/Hardware ISA is its execution substrate and its analog veto; and the critical review supplies the verification obligations that turn both from claims into a defensible chain.

8.3 The consolidated architecture, link by link

The chain has five links. For each, the table below names the synthesis’s *demand*, the project artifact that *supplies* it, and the residual *obligation*.

Table 8.1: Link-by-link consolidation

Link	Synthesis demand	Project artifact that supplies it	Residual obligation
Epistemic	A single qualified scalar φ , bounded, saturating, erring only toward higher risk	Gasification invariants ξ, S, set on the open-water conservation manifold	One-sided-bound proof; deterministic surrogate for set
Numerical	φ must not round in the optimistic direction	FP64 datapath + Posit-style format discipline; review’s rounding fix	Format-level proof of safe-side rounding
Control	Antitone admissible-command sets; firmly non-expansive projection; rate-clamp last	Monotonicity of S/ξ ; pole-zero antitone structure; <code>SFSPU.PROJ</code>	Exactness theorem beyond box constraints
Physical	Hardwired veto, bounded latency, fail-closed	<code>CONS.CHK</code> (2.7 ns), <code>TMR.VOTE</code> , <code>SAFE.ROLL</code> (28.4 ns), <code>msafety</code> CSR	Silicon realization vs. notional spec
Integrating	Conjunctive fail-closed release gate with provenance	Event registers (0=violated/1=satisfied), <code>ConstraintViolation</code> trap, write-ahead log	Liveness/feasibility proof

8.3.1 Epistemic link — the trusted scalar, instantiated

The synthesis’s Contribution I requires a scalar φ satisfying a five-part contract. The gasification invariants discharge most of this contract on inspection. $S \in \{0, 1\}$ is bounded and saturating by definition. ξ is bounded $O(1)$ by log-normalization against a geometric-mean baseline, despite raw gain varying across thirteen orders of magnitude — exactly the “scale-free” property a trusted scalar needs to compare regimes. Crucially, both are **monotone** with the observable they govern (yield, conversion), which is the property the actuation algebra consumes. And because they are read from a transfer-function topology that has already enforced conservation, they cannot, by construction, certify a state that violates the conserved quantity — the analogue of “feeding only constraint-tightening maps.”

What the gasification work does *not* yet provide — and the synthesis correctly identifies as the contribution most in need of grounding — is the *one-sidedness*: a proof that the scalar errs only toward higher risk. In the thermochemical setting this was never required, because the goal was accuracy, not conservatism. Re-deploying the invariant machinery as a safety gate means re-deriving each invariant as a one-sided bound: φ must lower-bound the true barrier margin under stated assumptions. This is the single largest piece of new mathematics the program owes, and it is now sharply scoped because the estimator itself already exists and is validated.

8.3.2 Numerical link — honesty of the scalar

A trusted scalar is only as honest as the arithmetic that computes it. The synthesis’s substrate chapter makes the format a safety variable: IEEE-754 single precision can drive an extended Kalman filter to diverge as its covariance loses positive-definiteness, while Posit arithmetic with

quire accumulation tracks a double-precision reference across the horizon. The PICAPD datapath is FP64 throughout and computes $K(m)$ by AGM, which reaches machine precision in ~ 5 iterations with error decaying as 2^{-2^n} — a numerically benign path.

The critical review supplies the discipline that turns “precise” into “safe-side.” Its remediation of the set claim — replace a wall-clock measurement with a **deterministic surrogate** such as the Kendall- τ inversion count, and demonstrate that ξ and S correlations survive the substitution — is exactly the form a numerical-honesty argument must take for any safety scalar: the computation must be reproducible and its rounding must be bounded on the conservative side. The obligation, then, is to (a) adopt the deterministic surrogate the review already specifies, and (b) prove the rounding mode of the φ datapath cannot move the reported margin in the optimistic direction.

8.3.3 Control link — the risk-monotone actuation algebra

Contribution II asks for a family of admissible-command sets indexed by a normalized risk band, **antitone** (rising risk only shrinks the set), each set a closed convex axis-aligned box so that projecting a proposed command onto it is firmly non-expansive (1-Lipschitz), implemented as a two-stage clamp — magnitude then rate, **rate-clamp last** — so that a risk spike produces bounded jerk rather than an actuator step-lurch. Multiple envelopes compose by **intersection (meet)**, **never union or average**, with **asymmetric hysteresis** (tighten instantly, loosen only after a dwell below a lower threshold).

Two project facts make this more than a proposal. First, the monotonicity the algebra depends on is *the same monotonicity the gasification work observed empirically*: S and ξ vary monotonically with the governed observable, “eliminating local optima” — which is precisely the antitone indexing the algebra needs to be well-posed. Second, the projection operator is already an instruction: **SFSPU.PROJ** performs constraint-manifold projection, and **CJACOB** supplies the Jacobian for constraint-manifold work. The “no averaging” prohibition is the safety counterpart of the gasification framework’s refusal to enforce conservation as a *soft penalty*; in both cases, taking a less-conservative blend destroys the worst-case bound.

The owed mathematics here is the **exactness theorem** — that sequential magnitude-then-rate clamping equals the true Euclidean projection. This holds for axis-aligned boxes but not for the coupled steering/acceleration limits and non-box friction circles of a real vehicle. The defensible path, consistent with the review’s general posture of downgrading overclaims to exactly-what-is-proven, is to prove the box approximation is itself conservative (its projection lies inside the true admissible set) and restrict the exactness claim to that regime until the non-box extension is proven.

8.3.4 Physical link — the timing-contracted analog veto

Contribution III is the link the synthesis treats most cautiously, because it invokes “PICAPD” as notional hardware. The project shows the hardware is specified, not notional — though still to be built. The veto is realized by the Safety/Constraint instruction class running beneath the software stack on the privileged path:

- **CONS.CHK** verifies $|\sum \text{inputs} - \sum \text{outputs}| < \text{tolerance}$ in **2.7 ns**, setting a conservation status bit; on failure the chain refuses to certify the command.
- **TMR.VOTE** takes the majority of three redundant decision channels — single-fault tolerant, and when two agree and one differs it flags the third as suspect (Byzantine detection).
- **SAFE.ROLL** restores the last known-good state from the write-ahead log in **28.4 ns**, the fail-closed default.

- The privileged `msafety` CSR (0x807) and the `ConstraintViolation` trap (code 9) provide the privilege isolation and architectural rollback the synthesis lists as required veto properties.

The timing contract the synthesis demands — *intervention latency* $\ll$ *sensor-consistency window* $\ll$ *digital decision window* — is satisfiable with these numbers against the documented sensor cadence (the slowest safety-relevant stream, LiDAR, completes a scan in 100 ms; `SFSPU.SYNC` reconciles streams in $< 10\mu\text{s}$; the veto fires in tens of nanoseconds). The honest framing the synthesis insists on still holds: these are *specified* latencies for a *requirements-complete* ISA, not measured properties of a fabricated die. FPGA-accelerated, lockstep-RISC-V realization remains the credible near-term substrate, and the program must not present cycle counts as measured silicon. But “notional” understates what exists: the contract, the instructions, the privilege model, and the latency budget are all written down.

8.3.5 Integrating link — the conformance gate, mechanized

Contribution IV composes the links into a single conjunctive, fail-closed release gate: a command may drive actuation only if, *simultaneously*, the trusted scalar is qualified, every fused input passed a pre-fusion admissibility test with provenance, tracking error is within its recurrent bound, the command lies in the intersection of active envelopes (rate-clamp last), no averaging or union appears on any safety path, the retrieval respected its budget, no online update silently revised ground truth, and any failed check has already forced the minimal-risk envelope.

The ISA mechanizes the conjunction directly. The 512 single-bit **event registers** encode constraint states with the safety-correct polarity (0 = violated, 1 = satisfied); the gate is the AND of the safety-bearing event bits. `RES.BUDG` discharges the compute/latency-budget clause. The **write-ahead log** — with per-entry timestamp, vector clock, transaction ID, and SHA-256 checksum — supplies the provenance and the “challengeable event” record. The `ConstraintViolation` trap makes *the absence of a discharged guarantee* itself a fault, which is exactly the synthesis’s conjunctive semantics (“the absence of a discharged guarantee is a fault”). What the corpus does not yet supply is the **liveness** proof — that the conjunction is not so strict it parks the vehicle forever — which §6 treats as the central open question.

8.3.6 Governance overlay — the slower assurance layer

The synthesis’s learning-theory bridge governs a policy that updates online: a Local Learning Coefficient (LLC) jump coupled with a regret drop signals a developmental phase transition, and an anti-silent-drift rule logs or quarantines any update that alters the policy’s effective truth judgments. The project’s contribution here is the substrate: `BYZ.CONS`, the trust-score register bank (256×32 -bit reputation metrics), and the write-ahead log give the proposer/verifier/adjudicator separation a place to record and quarantine challengeable events. The timescale honesty the synthesis demands must be preserved: LLC estimation runs on slow-loop timescales and **cannot** prevent a hallucinated command in *real time* — the analog veto is the real-time guarantee; the learning governance quarantines a *capability* between drives.

8.4 Standards crosswalk, grounded in concrete instructions

The constraint chain converts compliance from a statistical claim about how rarely the stack errs into a structural claim about what an error can do. The crosswalk below maps each standard to the *specific project mechanism* that answers it.

The decisive claim the chain licenses: **even under severe algorithmic hallucination in perception or planning, the actuation-boundary forms force the vehicle into a provable Minimum Risk Condition** — and do so without violent braking at ODD transitions, because the same rate-clamp-last discipline that prevents lurching is what carries the vehicle into the more conservative envelope.

Table 8.2: Standards crosswalk with ISA grounding

Standard	What it demands	Project mechanism that answers it
ISO 26262 (ASIL)	Mitigate E/E faults; freedom from interference	TMR.VOTE (single-fault tolerant), lockstep substrate, <code>msafety</code> privilege isolation, FP64/Posit numerical resilience
ISO 21448 (SOTIF) / UL 4600	Bound the “unknown unsafe” with no hardware fault	Trusted scalar (ξ, S) on the conservation manifold + actuation algebra make unsafe commands unrepresentable, not merely improbable
IEEE 2846 / RSS / SFF	Reasonable assumptions; behavior across ODD transitions	Rate-clamp-last bounded-jerk entry into a stricter envelope; asymmetric hysteresis carries the vehicle smoothly into fog/rain RSS parameters
AEC-Q100 / ISO 16750-3	Thermomechanical and vibration survivability	MEMS noise/vibration floor sets the real lower bound on any safety threshold; format/precision choices preserve covariance positive-definiteness

8.5 Verification obligations (the critical review, folded in)

The review was written against the gasification paper, but every one of its strongest objections re-applies — sharpened — to the safety re-deployment, and each becomes a verification obligation on the consolidated chain.

1. **The trusted scalar must be a proven one-sided bound (the largest debt).** The invariants are validated as *accurate*; they are not yet proven *conservative*. Each φ must be re-derived to lower-bound the true safety margin under stated assumptions, with a numerical-format argument that its computation cannot round optimistically.
2. **Replace sct with a deterministic surrogate.** The review’s own remediation applies verbatim: a wall-clock sort time is hardware-, JIT-, cache-, and scheduler-dependent and is “the single most exposed claim.” Substitute the Kendall- τ inversion count (or sorted-sequence entropy), demonstrate cross-machine rank correlation above ~ 0.85 , and show ξ/S correlations are preserved. A safety scalar cannot rest on a non-reproducible measurement.
3. **Downgrade mechanism claims to association.** The review flags “sct bridges thermodynamics and kinetics” and “elliptic integrals capture essential system behavior” as overclaims; the safety analogue is to never present the conservation-manifold parameterization as *mechanistically necessary*. State it as an effective reduced-order parameterization that is monotone and conservative — which is all the actuation algebra requires.
4. **Owe the exactness theorem.** Prove sequential clamping equals Euclidean projection for the box regime and that the box is conservative; flag the coupled/non-box extension as open.
5. **Prove liveness, not just safety.** A scalar that always reports maximum risk satisfies every clause and parks the vehicle. The genuine research question — and the program’s central empirical claim — is the trade-off frontier between the worst-case bound and ordinary drivability, the same tension the robust-CBF adaptation literature already grapples with.

6. **Keep the notional/built boundary visible.** Present PICAPD latencies and opcodes as *required and specified* properties, not measured silicon; name FPGA/lockstep-RISC-V as the realizable substrate.
7. **Cut or flag unverified citations.** The “Learning Mechanics” umbrella and its associated preprint could not be confirmed; cite the underlying components (singular learning theory, the LLC, implicit-bias results, equivariance, mechanistic interpretability) individually.

8.6 Limitations and the build plan

The consolidation strengthens the synthesis’s standing — the contributions are no longer free-floating proposals — but it does not discharge the four open obligations the synthesis itself named, and it adds the gasification-specific debts above. Honestly stated, what remains to be built is:

- a **one-sidedness proof** for at least one trusted scalar on a defined safe set, with a matching format-level rounding argument;
- a **deterministic re-implementation** of the non-equilibrium proxy and a cross-machine reproducibility study;
- the **exactness theorem** for the box regime plus a conservatism argument for the box approximation;
- a **liveness/feasibility study** mapping the trade-off frontier between conservatism and drivability;
- an **FPGA realization** of the Safety/Constraint class demonstrating the documented latency budget under jitter bounds; and
- a **certification dossier** instantiating the §4 crosswalk against a defined operational design domain.

The sequencing is natural: the deterministic-surrogate and one-sidedness work unblocks the trusted scalar; the exactness theorem unblocks the actuation algebra; the FPGA realization unblocks the veto’s timing claim; and only then does the liveness study and certification dossier become meaningful.

Provenance ledger (extended)

Table 8.3: Extended provenance ledger

Claim or construct	Standing	Role in the consolidated solution
Open-water simplex transform; CCA/Fibonacci/neural reduction; AGM elliptic parameterization; transfer-function conservation enforcement	Validated in the gasification corpus ($R^2 > 0.78$)	The conservation-manifold compiler; the engine of the trusted scalar
Invariants ξ (continuous), S (discrete parity), sct sct as a fundamental property	Validated as accurate; not yet proven conservative Flagged by the critical review	Instantiate Contribution I; one-sidedness owed Must be reframed as a machine-conditioned proxy / replaced by a deterministic surrogate
PICAPD ISA: AGM.ITER, ELI.COMP, TXF.PASS, MOM.*, VERLET/HAMILT	Specified (requirements-complete)	Execution substrate for the compiler
PICAPD Safety class: CONS.CHK (2.7 ns), TMR.VOTE, SAFE.ROLL (28.4 ns), msafety CSR, ConstraintViolation trap	Specified with latencies; not fabricated	Instantiate Contributions III and IV; FPGA realization owed
Sensor-Fusion class SFSPU.* tuned to IMU/camera/LiDAR/GPS + EKF	Specified	Establishes the sensor-consistency window of the timing contract
Guarded Swarms analog-veto principle	Established (verified publication)	Conceptual basis of the veto
CBF / RoM-FoM / RTF / Poisson safety / neural & gameplay filters; SLT/LLC; ISO 26262 / SOTIF / UL 4600 / RSS / IEEE 2846	Established literature	Accepted foundations of the chain
“Trusted Scalar” contract; “Risk-Monotone Action Algebra”; exactness theorem	Proposed	Now instantiated by project artifacts; theorems still owed
“Constitutional Truth Governance” / “Universal Filtration Primitive”	Not in literature	Design metaphors for the proposer/verifier/adjudicator overlay only
“Learning Mechanics” coalition umbrella + associated preprint	Could not be verified	Cite underlying components individually

A Consolidated Program for Constitutionally Governed, Provably Bounded Automated Driving (L2–L5): Synthesis of the Project Corpus into a Single Buildable Solution

9.1 The goal, stated precisely

The corpus points at one objective, expressed twice in two vocabularies. The roadmap states it as: *build a constitutionally governed automated-driving stack whose shared primitives are sufficient for an immediate L2 product, whose authority regime can be reconstituted for L3, whose bounded-domain completeness can support real L4, and whose architecture remains compatible with L5 as SAE/NHTSA define it today.* The dissertation states it as: *make autonomous driving certifiable rather than merely tested — not by proving the stack never errs, but by proving that when it does, the error has nowhere unsafe to go.*

These are the same goal. The consolidated thesis of this document is that **they are achieved by the same artifact**, and that the project’s central error to date has been to develop them as separate programs.

Consolidated thesis. Build one vertical constraint chain. From the top it is an *authority law* — who holds the dynamic driving task and under what warrant. From the bottom it is a *consequence law* — the monotone contraction of the admissible-command set, enforced finally by physics. The two laws are not complementary additions; they are the same chain read in opposite directions, and they are bound together at the EPU. The EPU is simultaneously the lowest organ of constitutional authority and the highest layer of the actuation-boundary refusal architecture.

Everything below makes that identification load-bearing.

9.2 The seam: where the two architectures meet

The EPU/constitutional materials and the dissertation describe four shared commitments under different names. Making the synonymy explicit is the first act of consolidation, because it shows the join is not a bridge between two systems but a single set of requirements that both halves were independently converging on.

The right reading is that the EPU docs specified the *governance obligations* of these four requirements and the dissertation specified their *mechanical and physical discharge*. The EPU

Table 9.1: Synonymy of constitutional and dissertation formulations

Constitutional / EPU formulation	Dissertation formulation	The unified requirement
Verification is prior to fusion; admissibility = integrity, alignment, scope, replayable lineage	Pre-fusion admissibility test + provenance (checklist item 2); singular-MDL floor on lossy perception	No state governs action until its evidentiary warrant is discharged
Safety monotonicity: as warrant falls, behavioral risk must not rise	Risk-monotone (antitone) actuation algebra: rising risk only shrinks the admissible set	The admissible-command set may only contract as one moves toward actuation and as risk rises
No pathway is both sole proposer and sole judge; EPU is the independent verifier	Proposer / verifier / adjudicator separation; analog veto with no programmable logic in the guard path	Independence between proposal and canonization, enforced beneath software
Anti-silent-drift release governance; training/inference separation	Anti-silent-drift rule keyed to LLC/stagewise phase-transition detection	Capability may improve; reliance conditions may not mutate invisibly

was described as a “deterministic side channel” that “validates commands” and “triggers MRC” — but the documents never said, in defensible terms, *what the validation computes* or *why the override is trustworthy beneath a compromised stack*. The dissertation answers exactly those two questions: the validation computes a one-sided **trusted scalar** and projects commands onto a **risk-monotone envelope**; the override is trustworthy because it is a hardwired analog veto under an explicit **timing contract**, fail-closed by construction.

Conversely, the dissertation never sequenced its guarantees against the SAE authority levels, never produced a commercial entry path, and never reconciled its established/proposed/unverified ledger with a deployment program. The EPU roadmap supplies all three.

The consolidation is therefore exact: **the dissertation is the formal specification of the EPU; the EPU roadmap is the deployment program for the dissertation.**

9.3 The unified stack

The consolidated system is a single vertical stack with two governing directions of flow. Authority flows downward (the user delegates to the mission layer, which delegates bounded execution to the stewards, which admit intent to the controller); consequence-bounding flows upward as a guarantee (the analog veto bounds what the controller can do, which bounds what the steward can authorize, which bounds what the mission layer can promise the user). The layers, top to bottom:

Mission layer (user as principal). Route, engagement, automation mode, broad style, and — critically — *silence as a standing-preference operator*. This is unchanged from the EPU docs and needs no addition from the dissertation; it is purely an authority concern with no direct actuation consequence.

Steward layer (delegated admissibility). Admit / modify / deny under the fixed hierarchy **safety** $\succ$ **law** $\succ$ **system integrity** $\succ$ **user preference**. The dissertation contributes one discipline the EPU docs lacked: the steward’s denial reasons must compose by **conservative**

join only — never average, never less-conservative-meet. When two admissibility judgments conflict, the system takes the tighter, not the mean. This closes a latent governance hole: a steward layer that “balanced” a safety constraint against a preference constraint would silently violate monotonicity.

Maneuver / controller layer. Translates admitted intent into dynamically feasible motion. Acts only on admitted intent; never redefines the admissibility boundary.

Reflex / EPU layer — the consolidated core. This is where the two architectures fuse. The EPU is redefined as the layer that computes and enforces the dissertation’s contributions in hardware-bounded form:

1. *Trusted scalar (φ).* The EPU’s admissibility check is no longer an unspecified “gate.” It is the computation of a qualified scalar that satisfies the five-part contract — denotes a genuine barrier-margin quantity, saturates, **errs only toward higher risk**, feeds only constraint-tightening maps, and re-qualifies continuously, escalating monotonically to maximum risk on any conformance failure. This is the rigorous form of “verification before fusion”: fusion is admitted only if φ lower-bounds the true safety margin.
2. *Risk-monotone projection.* The EPU’s command validation is the projection of the proposed actuator command onto the intersection of all active envelopes — closed, convex, axis-aligned boxes indexed by risk band — using the firmly non-expansive (1-Lipschitz) two-stage clamp with **rate clamp applied last**, so a risk spike produces bounded jerk rather than a step-lurch. This is the rigorous form of “safety monotonicity under degraded evidence.”
3. *Analog veto under timing contract.* The EPU’s “side-channel independence” and “route-to-MRC” become the dissertation’s hardwired guard: intervention latency $\ll$ sensor-consistency window $\ll$ digital decision window, no programmable logic in the guard path, **fail-closed** (maximal safe braking, zero positive acceleration) on corrupted data, missed deadline, or tampered signal. This is the rigorous form of “no pathway is both sole proposer and sole judge.”

The EPU thus stops being “a backup chauffeur” (a framing both the L2 and L3 docs explicitly rejected but could not fully replace) and becomes what the dissertation calls the *architecture of refusal*: the organ that makes unsafe commands physically unrepresentable.

Numerical substrate (cross-cutting). The dissertation’s most easily overlooked contribution belongs here: any scalar that gates actuation must be computed in a format whose rounding error is bounded on the *safe* side. IEEE-754 single precision can drive a Kalman covariance to lose positive-definiteness over a few hundred seconds; Posit-with-quire tracks a double-precision reference. φ and the envelope bounds must be computed in arithmetic that cannot round optimistically. The EPU docs never addressed numerics; without this, the trusted scalar’s one-sided guarantee is aspirational regardless of the algebra above it.

9.4 Authority regimes across L2–L5 on the unified stack

The roadmap’s strongest instinct — that L2/L3/L4/L5 are *different authority regimes on a shared substrate, not performance tiers* — survives consolidation intact and is strengthened, because the dissertation’s monotonicity guarantee is the one invariant that makes the substrate genuinely shared. The same analog veto and the same risk-monotone algebra hold at every level; only the *authority law above them* changes.

L2 — driver-supervised, commercial entry. Human holds the dynamic driving task. The unified stack runs with the EPU enforcing φ and the envelope, but the minimal-risk fallback resolves to *takeover-with-supervision-preserved* rather than autonomous MRC. L2 is the first field proof that the actuation-boundary refusal architecture works under production pressure — and, decisively, it is where the dissertation’s open **liveness problem** is first confronted commercially: a φ that always reports maximum risk is vacuously safe and forbids ordinary driving. L2’s validation program is the cheapest place to map the worst-case-bound-versus-feasibility frontier.

L3 — conditional automation, authority reconstitution. The machine holds the dynamic driving task inside a declared ODD; the human is fallback-on-request. This is the first genuine authority transfer, and the dissertation’s contribution becomes safety-critical rather than merely assistive: when the machine is the driver, the guarantee that *a hallucinated command has nowhere unsafe to go* is the entire safety case for the automated interval. The EPU’s route-to-MRC ceases to be a backup and becomes the constitutional fallback of a system that is temporarily the principal executor. L3 is not a software uplift from L2; it is a new authority law over an unchanged consequence law.

L4 — bounded autonomy, no engaged fallback inside the domain. The machine owns continuity, failure management, and safe-stop authority inside the ODD. The actuation-boundary forms now carry the full weight: there is no human to repair an inadmissible command in real time, so the fail-closed analog default *is* the safety net. Strategy is unchanged from the roadmap — complete one or two bounded envelopes (freight corridors, mapped highway lanes, geo-fenced logistics) operationally before generalizing.

L5 — universalization. The dissertation is candid where the roadmap is candid: this is the standards ceiling, not the near-term target. The consolidated view sharpens *why*. The analog veto and risk-monotone algebra remain necessary at L5 but become insufficient alone, for a reason the dissertation names precisely: the “**price of implicit bias.**” A policy that generalizes beautifully on clean data can be systematically fragile in exactly the direction a rare scene or adversary exploits, because the optimizer’s geometric bias is misaligned with the threat geometry. Bounded ODDs hide this; open-world operation exposes it. L5 therefore requires elevating *alignment between optimizer bias and safety threat geometry* to a first-class design constraint — a research obligation, not a deployment milestone.

The hardest pivots, as the roadmap argued, are **L2→L3** (authority transfer) and **L4→L5** (universalization). The consolidation adds that these are hard for *different* reasons: L2→L3 is an authority-law problem the EPU governance solves; L4→L5 is a consequence-law problem the dissertation’s open obligations expose.

9.5 Training/inference separation reconciled with the evidentiary ledger

Both architectures insist on separating an off-board training substrate from an on-board inference substrate. The dissertation adds a second separation the EPU docs did not formalize: the line between **what is established, what is proposed, and what could not be verified**. Consolidation requires holding both separations at once, because they govern different risks — the first prevents silent capability drift into the field; the second prevents unproven mechanism from masquerading as discharged guarantee.

Inference substrate (on-board). Executes only signed, frozen artifacts under the MB/TB/Gate allocation — Tailored Brain carries routine policy mass at low latency, Gate escalates to Main-Brain-derived capacity only when expected regret exceeds the latency/compute burden, Main Brain supplies tail-case competence off-board. The dissertation’s contributions

live *below* this as the EPU’s enforcement layer: the policy proposes, the EPU refuses what falls outside the monotone envelope. Crucially, the analog veto operates in microseconds; the cognition allocation in milliseconds.

Training substrate (off-board). Builds and validates MB/TB/Gate, mines failures through a structurally independent challenger, and packages signed releases with lineage, scope, and roll-back. The dissertation’s learning-theory bridge plugs in here: the **LLC / stagewise-transition detector** gives an online-estimable signal that a policy has acquired a new capability (a coupled regret-drop / LLC-jump), feeding the **anti-silent-drift** rule — any update altering effective truth judgments is logged as a challengeable event and quarantined where warranted.

The timescale honesty that consolidation must preserve. The dissertation’s most important caution is that these two layers operate on incommensurable clocks. The analog veto is the *real-time* guarantee (microseconds); the learning governance is a *slower assurance layer* (training-time / between-drives). The detector therefore **cannot** prevent a hallucinated command in real time — it can only quarantine a *capability* between drives. Conflating the two would overstate the result. The consolidated claim must be stated exactly: the analog veto bounds the consequence of any single command now; the learning governance bounds the drift of the policy over releases. Neither substitutes for the other.

The provenance ledger travels with the program. The dissertation’s appendix is not documentation hygiene; it is a deployment artifact. The consolidated program inherits its discipline directly: CBF/RTF layered control, Poisson and neural/gameplay safety filters, the sensing/numerics substrate, SLT/LLC instrumentation, the standards landscape, and the Guarded-Swarms analog-veto principle are **established** and cited as foundation. The trusted scalar, the risk-monotone algebra, the constitutional governance framing, and the EPU’s “constitutional enforcement” identity are **proposed** — the program’s own contributions, to be defended, not cited as prior art. The “PICAPD ISA” with its specific die/register/CSR figures, and the “Learning Mechanics” umbrella, are **unverified** — to be treated as notional requirements sketches and removed from any load-bearing path, with FPGA-accelerated kernels and lockstep RISC-V named as the realizable substrate. Carrying this seam unblurred is itself a certification asset.

9.6 The integrated conformance gate

The L2 and L3 rewrites each specified a validation burden (timing bounds, transform coherence, admit/modify/deny branching, MRC triggers, false-admission/false-denial rates, degraded-mode conservatism, interface-governance honesty). The dissertation specified an eight-item fail-closed release gate. Consolidation fuses them into one conjunctive gate, where the *absence* of a discharged guarantee is itself a fault. A reconstruction may drive actuation only if, simultaneously:

1. φ is a qualified trusted scalar, computed in safe-side-bounded arithmetic;
2. every fused input passed pre-fusion admissibility and carries replayable provenance;
3. the reduced-order→full-order tracking error is within its recurrent bound (RTF);
4. the command lies in, or is non-expansively projected (rate-clamp-last) into, the **intersection** of all active risk-banded envelopes;
5. no averaging, union, or less-conservative-meet appears anywhere on a safety-bearing path — including in steward denial composition;
6. the cognition allocation (TB/Gate/MB) respected its compute/latency budget, with WCET enforced;

7. no online policy update silently revised ground truth (anti-silent-drift, LLC-detector clear);
8. any failed check has already placed the system in the minimal-risk envelope (fail-closed).

Items 1–4 and 8 are the dissertation’s actuation-boundary mechanics; items 2, 5, 7 are the constitutional governance laws; item 6 is the EPU/control-stack budget discipline. The gate is the single artifact that turns a collection of mechanisms into the claim the program is actually making: *the unknown unsafe is structurally bounded, not empirically sampled*.

9.7 The phased program with explicit research obligations

The roadmap’s phase structure is sound and is retained, with the dissertation’s open obligations attached as gating research questions rather than left implicit.

1. **Phase 0 — Foundation.** Build the universal substrate: the four constitutional laws, proposer/verifier separation, the MB/TB/Gate training architecture, the EPU role *as the dissertation specifies it*, and the integrated conformance gate. **Owed before exit:** the trusted-scalar one-sided-bound proof for the chosen safe set, and the safe-side numerical-format argument. Without both, φ is aspirational.
2. **Phase 1 — L2 productization.** Ship driver-supervised highway/structured-road operation. **Owed:** the first empirical mapping of the **liveness/feasibility frontier** — proof that the gate is conservative enough to be safe yet live enough to permit ordinary driving. This is the program’s central empirical question and it is open; L2 is the cheapest venue to attack it.
3. **Phase 2 — L3 reconstitution.** Bounded highway conditional automation with takeover doctrine and machine-held route-to-MRC. **Owed:** the **exactness theorem beyond box constraints** — the sequential magnitude-then-rate clamp equals Euclidean projection only for axis-aligned convex boxes, but real actuator limits (coupled steering/acceleration, friction circles) are not axis-aligned. Either extend the theorem to the relevant non-box geometry or restrict claims to the regime where the box approximation is itself conservative. This must be discharged before the machine becomes the principal executor.
4. **Phase 3 — L4 domain completion.** One bounded domain, no engaged fallback inside it, machine-owned safe stop. **Owed:** demonstration that the fail-closed analog default is sufficient as the sole real-time safety net inside the ODD, given the timescale separation of §5.
5. **Phase 4 — L4 expansion.** Several domains on the same substrate; domain-transfer playbook; fleet economics.
6. **Phase 5 — L5 research program.** Kept as a frontier, not a product promise. **Owed:** treating optimizer-bias/threat-geometry alignment as a first-class constraint, plus the open-world generalization, adverse-weather, and cross-domain-transfer studies. No false claims of L5; continuous movement toward universalization without contaminating the near-term business.

9.8 Why this is the right level of abstraction

A monolithic end-to-end stack blurs every boundary the program depends on at once: it cannot separate proposal from verification, cannot keep evidence-admissibility distinct from planner preference, cannot keep training-time evolution out of field authority, and offers no place to anchor a physical guarantee. The consolidated architecture refuses that blur deliberately. The

lab learns aggressively; the road executes conservatively; and beneath the road, an analog layer enforces the narrowing with electricity rather than code.

The payoff is a certification posture road-miles alone cannot reach. Against ISO 26262 it offers hardware veto, bounded latency, lockstep substrate, and numerical resilience. Against SOTIF/UL 4600 it converts compliance from a statistical claim about *how rarely* the stack errs into a structural claim about *what an error can do*. Against IEEE 2846 / RSS it enters stricter envelopes at bounded jerk via rate-clamp-last, so the same discipline that prevents a lurch is what carries the vehicle smoothly into the more conservative parameters that fog or rain demand.

9.9 Final formulation

Build one constraint chain that is an authority law from the top and a consequence law from the bottom, joined at the EPU. Above the EPU, a constitutional order governs *who* may decide: the user is principal, the stewards hold bounded admit/modify/deny authority under a fixed hierarchy, learning occurs only off-board, and reliance conditions never mutate invisibly. At and below the EPU, a risk-monotone actuation algebra governs *what an error may do*: a trusted scalar that errs only toward higher risk gates fusion, a non-expansive rate-clamp-last projection contracts the admissible-command set as risk rises, and a hardwired fail-closed analog veto under an explicit timing contract makes unsafe commands physically unrepresentable beneath any software fault. The same chain carries four authority regimes — supervised L2, conditional L3, bounded L4, universal L5 — by changing only the law above the EPU while the consequence law below it stays invariant. The program ships L2 first to confront the liveness frontier cheaply, reconstitutes authority for L3 once the exactness theorem is discharged, completes bounded L4 where the fail-closed default is the sole real-time net, and treats L5 as a research frontier where optimizer-bias/threat-geometry alignment becomes first-class. Every load-bearing claim carries its standing — established, proposed, or unverified — and the seam between what is known and what is claimed is never blurred. The wager is not that the stack never errs, but that when it does, the error has nowhere unsafe to go.

Appendix — What changed in consolidation

Table 9.2: Consolidation changes

Before consolidation	After consolidation
EPU described as “deterministic side channel” / “gates,” mechanism unspecified	EPU = trusted-scalar computation + risk-monotone projection + timing-contracted analog veto
“Safety monotonicity” stated as a principle	Stated as the antitone actuation algebra with a non-expansive, rate-clamp-last projection and an owed exactness theorem
Levels framed as authority regimes on a shared substrate	The shared substrate is <i>specifically</i> the invariant consequence law; only the authority law above the EPU changes per level
Training/inference separation as engineering hygiene	Joined to the LLC/stagewise detector and to an explicit real-time-veto / between-drives-quarantine timescale separation
Validation burdens listed per level (L2, L3)	Fused into one conjunctive fail-closed conformance gate where a missing guarantee is itself a fault
Numerics unaddressed	Safe-side-bounded arithmetic (Posit/quire) made a precondition of the trusted scalar
Evidentiary standing implicit	Established/proposed/unverified ledger carried as a deployment artifact; PICAPD and “Learning Mechanics” downgraded to notional/removed

A Solution Architecture: Provably Bounded Autonomous Driving

Purpose

This document consolidates the findings distributed across the project corpus — the safety-stack control theory, the automotive sensing and numerics survey, the learning-dynamics theory, the elliptic-filter design mathematics, and the structure-preserving simulation methods — into a single buildable solution. The goal is the one stated by the dissertation synthesis: engineer safety not as accuracy but as *an architecture of refusal*, in which every degree of freedom available to a fallible perception-and-planning stack is narrowed, stage by stage, until an unsafe actuator command is rendered physically unavailable at the actuator interface.

Evidentiary discipline

The solution separates what is **established** (load-bearing prior art accepted as foundation) from what is **proposed** (the original synthesis, advanced as engineering hypotheses with explicit verification obligations). That seam is never blurred. A provenance ledger closes the document.

10.1 The problem the solution must solve

Autonomous vehicles fail in a characteristic way: the hardware functions perfectly while the algorithm misreads a novel scene, mis-fuses a corrupted sensor stream, or hallucinates under a learned policy whose behavior on the present input was never directly tested. Classical functional safety (ISO 26262) was built for the opposite failure — it assumes the *hardware* breaks and the specification is correct. The gap between these two pictures is what ISO 21448 (SOTIF) names “**the unknown unsafe**,” and it is precisely the territory in which a system like a city-driving Level 2+ stack actually operates.

The conventional response is to make perception and planning more accurate. The solution advanced here rests on the opposite premise: **accuracy is the wrong target**. A stack can be made arbitrarily accurate on average and still emit a catastrophic command on the one input that matters. The solution therefore makes the *consequences* of error bounded by construction — accepting that the high-level stack is fallible, and interposing a chain of constraints (epistemic, numerical, control-theoretic, and finally electrical) so that the worst an error can do shrinks monotonically as it travels toward the wheels.

10.1.1 The solution thesis, stated as a monotonicity property

As one moves from perception toward actuation, and as estimated risk rises, the set of actuator commands the system is permitted to authorize may only contract, never expand — and at the final, pre-actuation stage this contraction is enforced by physics rather than by software.

Everything in this document is an attempt to make that property precise, ground each link in established theory, and identify where a link is currently a proposal rather than a proof. The result is a five-stage pipeline in which each stage hands the next a strictly tighter envelope of admissible commands.

10.2 Solution overview — the constraint chain

The solution is a single pipeline of five stages. Each stage is a **constraint-tightening map**: it can only narrow what the next stage is allowed to do. The pipeline terminates in an analog layer that enforces the narrowing with electricity, beneath any software fault.

Table 10.1: Five-stage constraint chain

Stage	Name	What it guarantees	Established foundation it rests on
S0	Trusted sensing & numerics	Every number feeding a safety margin is honest, with rounding bounded on the safe side.	MEMS noise floors; Posit vs IEEE-754 in the EKF; symplectic prediction; sensor redundancy.
S1	Fidelity gate (trusted scalar φ)	Discarded perceptual detail can only shrink, never enlarge, the admissible action set.	Barrier-margin witnesses; singular-MDL floor on lossy compression.
S2	Layered safety filter	A cheap guarantee proven on a reduced model projects onto the physical vehicle.	Control-barrier functions; recurrent tracking functions; Poisson safety; neural & gameplay filters.
S3	Risk-monotone actuation algebra	Rising risk only shrinks the command box; projection cannot amplify oscillation.	Convex projection (1-Lipschitz); RSS / SFF / IEEE 2846 duty-of-care envelopes.
S4	Timing-contracted analog veto	A hallucinating or compromised stack physically cannot reach the actuator first.	Guarded-Swarms analog-veto principle; RISC-V dual-core lockstep to ASIL-D.

Reading the chain. S0 and S2 are drawn almost entirely from established work and are accepted as foundation. S1, S3, and the on-road adaptation in S4 are the project’s *proposed* contributions — novel syntheses of that foundation, advanced as hypotheses to be defended. Section 7 states each proposal’s open verification obligation plainly.

10.3 Stage-by-stage: the established foundation

This section records the load-bearing prior art the solution builds on. None of it is original; all of it is verified in the corpus.

10.3.1 S0 — Trusted sensing and numerics as the substrate

A safety margin is only as honest as the numbers feeding it. Three established facts set the floor of the entire chain:

- **The numerical format is a safety variable.** Standard IEEE-754 single precision can drive an extended Kalman filter to diverge after several hundred seconds as the covariance loses positive-definiteness; Posit arithmetic with quire accumulation tracks a double-precision

reference across the full horizon. Any scalar that gates actuation must be computed in a format whose rounding error is bounded on the safe side.

- **Sensor noise sets a hard threshold floor.** MEMS inertial noise floors and vibration-rectification offsets place a real lower bound on how small a safety threshold can be; this is what AEC-Q100 / ISO 16750-3 thermomechanical and vibration qualification ultimately constrains.
- **Prediction must not inject energy.** Symplectic integrators (Størmer / Velocity-Verlet) bound the simulated energy to oscillate about its true value with global error of order $O(\Delta t^2)$, avoiding the unphysical “teleportation” of non-symplectic schemes — the same structure-preserving discipline that lets exascale plasma PIC codes remain trustworthy over millions of timesteps.

Grounding in the product reality. A shipping city-driving stack already instantiates the redundancy this stage demands: a sensor suite of 10 cameras, 5 radars, and 12 ultrasonics gives a physics-diverse failure surface rather than a single modality. Near-field ultrasonics provide hard distance limits at the bumpers where camera depth degrades and radar returns become unstable; their $\sim 10\text{--}20$ ms sensing cycle and centimetre-class resolution make them the natural source of the “hard limits” that the actuation algebra later enforces. This is robustness over minimalism — exactly the substrate S0 requires.

10.3.2 S2 — The layered safety filter

A high-dimensional vehicle cannot have control-barrier functions (CBFs) synthesized on its full dynamics under microsecond deadlines. The accepted resolution is **layered control**: synthesize the CBF on a low-dimensional reduced-order model (RoM) that captures the safety-critical kinematics, and task a low-level controller with forcing the full-order model (FoM) to track it. The guarantee is only as good as the tracking bound.

- **Recurrent tracking functions (RTFs)** relax strict monotonic Lyapunov decay to a weaker finite-time recurrence condition, permitting transient tracking error provided the trajectory returns to a recurrent safe set before leaving the safe region. RTFs are what let a guarantee proven cheaply on the RoM project onto the physical FoM.
- **Poisson safety functions** merge many constraints into a single smooth scalar field (solving Poisson’s equation with Dirichlet boundary conditions on the safe/unsafe sets) that resolves quickly.
- **Neural CBFs and gameplay filters** internalise these constraints offline via adversarial self-play between a safety-seeking policy and a worst-case-uncertainty adversary, giving zero-shot robustness without per-step worst-case over-tightening.

10.3.3 The standards landscape and the analog-veto principle

ISO 26262 governs electrical/electronic faults via the ASIL framework; SOTIF (ISO 21448) and UL 4600 address performance-limitation hazards with no hardware fault; RSS, SFF, rulebooks, and IEEE 2846 operationalise the behavioural duty of care. Tying the whole stack to ground is the **Guarded-Swarms principle** (a verified publication): digital coordination negotiates *what to do*, but independent analog guard channels at the actuator interface can veto an unsafe command within microseconds, with no programmable logic in the guard path. *Intelligence proposes; hardware disposes.*

10.3.4 The learning theory that makes the policy legible

Deep networks need no longer be treated as black boxes. Two established results are used as **instrumentation**, not decoration:

- **Singular learning theory** replaces parameter count with the Local Learning Coefficient (LLC) as the true measure of effective complexity, estimable at scale via preconditioned stochastic-gradient Langevin dynamics.
- **Stagewise training**: capabilities appear in discrete developmental phase transitions, visible as a sharp regret drop paired with an LLC jump. This gives a real, online-estimable signal that a learning policy has changed what it knows.

10.4 The proposed synthesis (original contributions)

This section advances the solution’s own claims. Each is a synthesis of the Section 4 foundations; none should be read as pre-existing literature. They are stated as proposals to be defended, with their anti-patterns and verification obligations made explicit. Where a specific hardware artifact is invoked it is treated as a *notional reference architecture* — a requirements sketch, never a citation to a built device.

10.4.1 Contribution I — S1 fidelity gate as a trusted scalar φ

Proposal. Reconstruction fidelity in the perception loop is not measured as reconstruction error (an L2 distance between input and rebuild). It is a single qualified scalar φ answering a different question: *can the discarded detail only ever shrink, never enlarge, the admissible action set?* φ must satisfy a five-part contract:

1. It denotes a genuine safety-relevant quantity — a barrier-margin or causal-sufficiency witness, not a PSNR proxy.
2. It is bounded and saturating.
3. **It overestimates risk whenever it errs.** A fast, fallible estimator becomes trustworthy not by being right but by being wrong only in the safe direction.
4. It feeds only constraint-tightening maps.
5. It is re-qualified continuously, escalating monotonically toward maximum risk on any conformance failure (trusted $\rightarrow$ suspect $\rightarrow$ maximal risk).

The inversion that makes it original. Classical estimation minimises average two-sided error. This proposal deliberately bounds worst-case *optimistic* error and compiles that bound into an additive conservative bias, so the reported margin globally lower-bounds the true margin.

Anti-patterns it forbids. Treating L2 accuracy as the safety criterion; letting a “high-fidelity” reading authorise a more aggressive action; and — critically — **averaging**. Combining a trusted scalar with an untrusted one, or taking the less-conservative of two, destroys the worst-case bound. The admissible algebra is restricted to the conservative join and safe-monotone composition.

10.4.2 Contribution II — S3 risk-monotone actuation algebra

Proposal. At the boundary immediately before actuation, define a family of admissible-command sets indexed by a normalised risk band, with the **antitone property** that rising risk only shrinks the set. Require every admissible set to be a closed, convex, axis-aligned

box, so that projecting any proposed command onto it is a unique, **firmly non-expansive (1-Lipschitz)** operation — guaranteeing the enforcer can never amplify an oscillation injected by corrupted software.

Clamp ordering is a safety property. Implement the projection as a two-stage clamp — magnitude, then rate — with the **rate clamp applied last**. Magnitude-last ordering forces an unbounded step when a risk spike shrinks the box below the current state; rate-last ordering ratchets the command into the tighter envelope at maximal bounded jerk, producing a graceful distance-decay toward the new envelope (a Band-4 transition bleeding steering and acceleration down over $\sim 1.25\text{--}1.47$ s) rather than an actuator step-lurch.

Composition and hysteresis. When multiple envelopes apply, the effective envelope is their **intersection (meet)** — never union or average. Band switching uses **asymmetric hysteresis**: tighten instantly on crossing an up-threshold; loosen only below a strictly lower down-threshold after a dwell time. Symmetric hysteresis is rejected because it delays tightening — the one direction that must never be delayed.

A note on shaping the envelopes. Where a band transition must shape a frequency-selective response (e.g. attenuating high-frequency command content above a cutoff while preserving low-frequency authority), the elliptic-filter design mathematics in the corpus supplies the optimal tool: for a given order, an elliptic (Cauer) response achieves the narrowest transition band of any filter, with the minimum order fixed analytically by ratios of complete elliptic integrals of the first kind. The accompanying caution is itself a safety fact — elliptic filters have non-linear phase and elevated group-delay variation, so any such shaping on a safety-bearing path must budget that delay explicitly against the timing contract of Stage S4.

10.4.3 Contribution III — S4 timing-contracted analog veto

Proposal (grounded in real prior art). The actuation algebra of S3 is enforced by an independent analog layer under an explicit timing contract:

$$\text{intervention latency} \ll \text{sensor-consistency window} \ll \text{digital decision window}.$$

Because the guard path is hardwired comparators and latches with no programmable logic, a compromised or hallucinating software stack cannot reach the actuator before the guard acts. On corrupted data, a missed deadline, or a tampered signal path, the guard forces the minimal-risk envelope (maximal safe braking, zero positive acceleration) — a **fail-closed** default rather than last-good-command coasting.

The notional hardware. Any conforming enforcer must exhibit: privileged, non-user-writable safety registers; deterministic bounded latency; and architectural rollback to a known-safe configuration on a violated safety bit. These are stated as *required properties*, not as a built chip. FPGA-accelerated kernels over a RISC-V dual-core-lockstep substrate (the established ASIL-D path) are the credible near-term realisation of the jitter-bounded reflexive layer.

10.4.4 Contribution IV — the conformance checklist as the integrating artifact

The contributions compose into a single **fail-closed release gate**. A reconstruction may drive actuation only if, simultaneously, all eight conditions hold. The items are conjunctive: the absence of a discharged guarantee is itself a fault.

1. φ is a qualified trusted scalar (satisfies the full five-part contract).
2. Every fused input passed a pre-fusion admissibility test and carries provenance.
3. The RoM→FoM tracking error is within its recurrent bound.
4. The command lies in (or is non-expansively projected into) the intersection of all active envelopes, rate-clamp last.

5. No averaging, union, or less-conservative meet appears anywhere on a safety-bearing path.
6. The retrieval that produced the reconstruction respected its compute / latency budget.
7. No online policy update silently revised ground truth (Section 6).
8. Any failed check has already placed the system in the minimal-risk envelope.

This checklist is the solution’s claim to systematicity — it is what turns a collection of mechanisms into an argument that the unknown unsafe is *structurally bounded rather than empirically sampled*.

10.5 Governing an updating policy

A vehicle whose policy updates online raises a governance problem the actuation layer cannot solve: the policy may quietly acquire a new behaviour. The solution joins the two halves of the corpus here.

Detector (established). The LLC and stagewise-transition results give a real, online-estimable signal: a developmental phase transition appears as a coupled regret-drop / LLC-jump. The singular-MDL result further bounds how far decomposition can compress an input before destroying causally necessary structure — a principled floor on lossy perception that feeds directly into the φ contract of Stage S1.

Governance (proposed). The overlay is an **anti-silent-drift rule**: any update that alters the policy’s effective truth judgments is recorded as a formal, challengeable event rather than absorbed into the canon. The detector fires; the governance logs and, where warranted, quarantines. This is an architectural pattern — proposer / verifier / adjudicator separation, with ambiguity preserved when evidence is insufficient — not an established governance theory.

A genuine open risk: implicit bias as a double-edged sword. The optimiser’s implicit geometric bias — the very thing that makes overparameterised networks generalise — can *de-grade* robustness when it is misaligned with the adversarial threat model’s geometry (the “price of implicit bias”). A network that generalises beautifully on clean data may be systematically fragile in exactly the direction a rare scene or attacker exploits. The solution treats alignment between optimiser bias and the safety threat geometry as a first-class design constraint, not an afterthought.

10.6 Certification argument and standards crosswalk

The payoff of the constraint chain is a certification posture that empirical road-miles alone cannot reach. The decisive claim is narrow but strong: *even under severe algorithmic hallucination in perception or planning, the actuation-boundary forms force the vehicle into a provable Minimum Risk Condition*.

This converts SOTIF compliance from a statistical claim about how rarely the stack errs into a structural claim about what an error can do — and it does so without violent braking at operating-domain transitions, because the same rate-clamp-last discipline that prevents lurching is what carries the vehicle smoothly into the more conservative RSS parameters that fog or rain demand.

10.7 Critical appraisal — what is owed, not yet proven

A solution is only honest if it states where it could fail. Each open obligation below is a concrete work item, not a caveat.

Table 10.2: Standards crosswalk

Standard	What it demands	Where the solution answers it
ISO 26262 (ASIL)	Mitigate E/E faults; freedom from interference.	Stage S4 hardware veto, deterministic bounded latency; lock-step substrate; S0 numerical resilience.
ISO 21448 / SOTIF; UL 4600	Bound the “unknown unsafe” with no hardware fault present.	S1 trusted-scalar fidelity gate + S3 actuation algebra make unsafe commands physically unrepresentable.
IEEE 2846 / RSS / SFF	Minimum reasonable assumptions; safe behaviour across domain transitions.	Rate-clamp-last bounded-jerk entry into a stricter RSS envelope; asymmetric hysteresis.
AEC-Q100 / ISO 16750-3	Thermomechanical and vibration survivability.	S0 vibration / noise floor sets the real lower bound on any safety threshold; format and precision choices.

10.7.1 The exactness theorem is owed

Contribution II’s claim that sequential magnitude-then-rate clamping equals Euclidean projection holds for axis-aligned convex boxes. Real actuator constraints (coupled steering / acceleration limits, non-box friction circles) are not axis-aligned. The work item: either prove the theorem extends to the relevant non-box geometry, or restrict the claim to the regime where the box approximation is itself conservative.

10.7.2 The trusted scalar can be vacuously safe — liveness is the central open question

A scalar that always reports maximum risk satisfies every clause of the φ contract and parks the vehicle forever. The hard part — unaddressed by the contract as stated — is liveness: proving the gate is not so conservative that it forbids ordinary driving. The genuine research problem is the trade-off frontier between the worst-case bound and feasibility, exactly the tension the robust-CBF adaptation literature already grapples with. This is the solution’s central empirical question, and it is open.

10.7.3 Detection latency versus actuation latency

The learning-theory detector (LLC estimation via SGLD) operates on training-time or slow-loop timescales; the analog veto operates in microseconds. The governance overlay therefore cannot prevent a hallucinated command in real time — it can only quarantine a capability between drives. The honest framing: the analog veto is the real-time guarantee; the learning governance is a slower assurance layer. Conflating their timescales would overstate the result.

10.7.4 The notional hardware must not masquerade as built

Specific cycle counts, opcodes, or register files must be presented as required properties (privilege isolation, bounded rollback latency, jitter-bounded execution), with FPGA / lockstep-RISC-V named as the realisable substrate — never as measured properties of an existing device.

10.7.5 Unverified citations must be removed or downgraded

Every citation on a load-bearing path must be independently verified before any external submission; unverifiable framing must be cut or explicitly flagged as the candidate’s own conjecture. The provenance ledger below is the working record of that discipline.

Provenance ledger

The integrity of the solution depends on never blurring the seam between what is known and what is claimed. This ledger is that seam, made explicit.

Closing statement

The contribution of this solution is not a new optimiser, sensor, or controller. It is *an architecture of refusal*: a disciplined chain in which each stage can only narrow what the next is permitted to do, terminating in an analog layer that enforces the narrowing with electricity rather than code. The foundations are real and cited as such; the integrating mechanisms are advanced as proposals with their open obligations stated plainly. The wager is that this is the right level of abstraction at which to make autonomous driving *certifiable* rather than merely *tested* — not by proving the stack never errs, but by proving that when it does, the error has nowhere unsafe to go.

Table 10.3: Final provenance ledger

Claim or construct	Standing	Treatment in this solution
CBF / RoM–FoM layered control / RTF / Poisson safety / neural & gameplay filters	Established	Accepted as foundation (Stage S2).
MEMS, DVS, quantum sensing, FFT accelerators, neuromorphic, RISC-V ASIL-D, Posit, Verlet, AEC-Q100	Established	Accepted as substrate (Stage S0).
Structure-preserving / symplectic integration (PIC, MHD), exascale fidelity over long horizons	Established	Accepted; energy-bounded prediction in S0.
Singular learning theory / LLC; stagewise phase transitions; implicit-bias results	Established	Accepted; LLC used as drift detector (Section 6).
Complete-elliptic-integral filter design (minimum order, Cauey optimality, phase trade-off)	Established	Accepted; optional envelope-shaping tool in S3, with group-delay budgeted against S4.
MB.Drive Assist Pro sensor suite (10 cameras, 5 radars, 12 ultrasonics; redundancy by design)	Established	Accepted as the concrete S0 substrate grounding the abstract chain.
ISO 26262, SOTIF/21448, UL 4600, RSS, SFF, IEEE 2846	Established	Accepted; certification crosswalk (Section 8).
Guarded-Swarms analog-veto principle (microsecond actuator vetoes, no logic in guard path)	Established	Accepted; basis of Stage S4.
“Trusted Scalar” five-part contract (φ)	Proposed	Contribution I (Stage S1); one-sided-bound proof + safe-rounding argument owed.
“Risk-Monotone Action Algebra” / clamp-ordering	Proposed	Contribution II (Stage S3); exactness theorem owed beyond box constraints.
“Constitutional Truth Governance” / “Universal Filtration Primitive”	Proposed	Used as design metaphors only (Section 6); not cited as theory.
“PICAPD” ISA (24-core die, 512 event registers, <code>msafety</code> CSR)	Unverified	Treated as a notional requirements sketch for S4, never as a built device.
“Learning Mechanics” coalition umbrella + associated preprint ID	Unverified	Downgraded; underlying components cited individually instead.